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MILITARY SPECIFICATION

DATA AND TESTS, ENGINEERING: CONTRACT REQUIREMENTS

FOR

GUIDED MISSILE TARGET SYSTEMS

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

TABLE OF CONTENTS

	PARAGRAPH
SCOPE	1
Scope	1.1
APPLICABLE DOCUMENTS	2
Effectivity of documents	2.1
Specifications	2.1.1
Standards	2.1.2
Publications	2.1.3
Other publications	2.2
Use of Specifications and Standards	2.3
REQUIREMENTS	3
Action	3.1
Data to be furnished	3.1.1
Quantity, type and distribution	3.2
Quantity	3.2.1
Type	3.2.2
Distribution	3.2.3
Reports	3.3
Engineering data required	3.4
Program master plan	3.4.1
Development program planning report	3.4.2
Reliability reports	3.4.3
Environmental criteria report	3.4.4
TARGET System Brochure	3.4.5
Program evaluation and review (PERT) reporting requirements	3.4.6
Monthly progress letters	3.4.7
Development check points	3.4.8
Presentation data	3.4.9
Slides, charts, graphs, etc.	3.4.9.1
Motion picture film	3.4.9.2

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	PARAGRAPH
Simulator program reports	3.4.10
Aircraft installation report	3.4.11
Ship installations report	3.4.12
Handling equipment report	3.4.13
Shipping containers report	3.4.14
Aerodynamic and performance data	3.4.15
Aerodynamic investigations	3.4.15.1
Aerodynamic and thermodynamic heating analysis report	3.4.15.2
Stability and control report	3.4.15.3
System dynamics report	3.4.15.4
Characteristics and performance data	3.4.15.5
Launching characteristics data	3.4.15.6
Structural design, analyses and test data	3.4.16
Electrical systems data	3.4.17
Information on non-standard parts	3.4.17.1
Nomenclature, nameplates and serial numbers	3.4.17.2
Schematic diagrams	3.4.17.3
Master wiring diagrams	3.4.17.4
Aircraft - Target Electrical system data	3.4.17.5
Electrical load analysis	3.4.17.6
Material and processes development and evaluation report	3.4.18
Magnesium alloy parts list	3.4.19
Finish specification	3.4.20
Heating and cooling report	3.4.21
Hydraulic system data	3.4.22
Pneumatic system data	3.4.23
Recovery system report	3.4.24
Human factors data	3.4.25
Electronic equipment data	3.4.26
Control and stabilization system report	3.4.27
Guidance system report	3.4.28
Maintenance and operational test and checkout equipment report	3.4.29
Flight test equipment report	3.4.30
Telemetering Electronic Design Data	3.4.31
Antenna data report	3.4.32
Frequency allocation data	3.4.33
Aircraft Armament characteristics report	3.4.34
Launcher installations report	3.4.35
Weight and balance data	3.4.36
Propulsion system data	3.4.37
Test procedure manual	3.4.38
Development tests reports	3.4.39
Experimental flight test report	3.4.40

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Check the source to verify that this is the current version before use.

	PARAGRAPH
Preflight planning report	3.4.40.1
Preflight detailed flight plan for each flight	3.4.40.2
Post flight firing letter report	3.4.40.3
Post flight evaluation report	3.4.40.4
Handbook of operation and service instructions with parts breakdown	3.4.41
Safety documentation	3.4.42
Specification and specification revision pages	3.4.43
Contractor prepared specifications	3.4.43.1
Specification revision pages	3.4.43.2
Logbook	3.4.44
Demonstration plan	3.4.45
Demonstration report	3.4.46
Alteration report	3.4.47
Microfilm	3.4.48
Preparation of engineering drawings	3.4.48.1
Interchangeability data	3.4.49
Radar reflectivity report	3.4.50
Mission planning and operational manual	3.4.51
Quality Assurance Documentation	3.4.52
Summary of design data	3.4.53
Tests required	3.5
Development tests	3.5.1
Reliability tests	3.5.2
Acceptance tests	3.5.3
Service tests	3.5.4
Witnessing of tests	3.5.5
QUALITY ASSURANCE PROVISIONS	4
Sampling	4.1
Examination and action	4.2
Examination	4.2.1
Summary of design data	4.2.1.1
Action	4.2.2
Completeness of design data and drawings	4.2.2.1
Revision of design data	4.2.2.2
Releases of drawings	4.2.2.3
Design data submitted for information	4.2.2.4
PREPARATION FOR DELIVERY	5
Design data submittal	5.1
Method of submittal	5.2

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	PARAGRAPH
NOTES	6
Responsibility for applicable specifications and publications	6.1
Duplication of data	6.2
Revision of data	6.3
Security classification	6.4
Use of this specification	6.5
Definitions	6.6
Differences over prototype	6.6.1
Abbreviations	6.6.2
Deviations	6.7

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MILITARY SPECIFICATION
DATA AND TESTS, ENGINEERING CONTRACT REQUIREMENTS FOR GUIDED MISSILE TARGET SYSTEMS

1. SCOPE.

1.1 SCOPE. - This specification contains the requirements for engineering data and tests to be furnished under contracts for guided missile target systems.

2. APPLICABLE DOCUMENTS.

2.1 EFFECTIVITY OF DOCUMENTS. - The following documents form a part of this specification. Unless otherwise indicated in the applicable addendum to this specification, the issue in effect on date of invitation for bids shall apply.

2.1.1 SPECIFICATIONS.

Military

MIL-S-901	Shock Tests, H. I.
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-D-1000/1	Drawings, Engineering and Associated Data
MIL-W-3947	Weight and Balance Control System for Guided Missiles and Space Launch
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental Electronic Equipment
MIL-H-5474	Handbooks and Breakdowns; General Preparation of
MIL-D-5480	Data, Engineering and Technical Reproduction Thereof
MIL-P-5518	Pneumatic System; Aircraft Design, Instalation and Data Requirements for
MIL-E-6051	Electrical-Electronic System, Compatibility and Interference Control
MIL-I-6181	Interference, Controlled Requirements, Aircraft Equipment

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MIL-E-7016	Electric Load and Power Source Capacity, Analysis of: Method for Aircraft and Missiles
MIL-F-7179	Finishes and Coatings: General Specification for Protection of Aerospace Weapons Systems, Structures and Parts
MIL-M-7700	Flight Manual
MIL-H-7937	Handbooks Operation and Service Instruction with Parts Breakdown for Target Drones
MIL-E-8189	Electronic Equipment, Guided Missiles, General Specification for
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-A-8868	Airplane Strength and Rigidity Ground Tests
MIL-A-8870	Airplane Strength and Rigidity Vibration, Flutter and Divergence
MIL-C-9878/1	Cards, Tabulating and Microfilm
MIL-I-16910	Interference Measurement, Radio, Methods and Limits, 14 Kilocycles to 1000 Megacycles
MIL-F-17874	Fuel Systems Aircraft, Installation and Test of
MIL-D-17984	Data Presentation Requirements, Installed Performance and Air Induction System
MIL-D-18136	Research and Engineering Report; Format and General Requirements
MIL-D-18243	Demonstration of Guided Missile Weapon Systems; General Specification for
MIL-D-18300	Design Data Requirements for Avionic Equipment
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-I-18802	Fuel and Oil Lines, Aircraft, Installation of
MIL-M-18828	Mockups: Construction of, for Target Drones and Guided Missiles
MIL-P-21018	Preparation of Standard Aircraft Characteristics Charts and Performance Data for Guided Missiles
MIL-H-22174	Human Factors Data for Aircraft and Missile System

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MIL-P-23189	Pert/Time and Pert/Cost Management Information System for Planning and Control
MIL-P-25062	Parachute Recovery Systems, Missile and Drone, General Requirement for Development of
MIL-W-25140	Weight and Balance Control Data (For Airplanes and Rotorcraft)
MIL-H-25475	Hydraulic Systems, Missile, Design, Installation, Tests and Data Requirements, General Requirements for
MIL-M-81273	Technical Manuals
WS-4613	Air Systems Command Weapon Specification for Safety Requirements, Minimum, for Aerial Targets

2.1.2 STANDARDS.

Military

MIL-STD-15	Electrical and Electronic Symbols
MIL-STD-100	Engineering Drawing Practices
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-176	Weight and Balance Data Reporting Forms for Guided Missiles and Space Launch Vehicles
MIL-STD-254	Weight and Balance Data Reporting Forms for Aircraft
MIL-STD-470	Maintainability Program Requirements
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-1304	Reliability Reports

2.1.3 PUBLICATIONS.

SPECIAL AERONAUTICAL REQUIREMENTS

SAR-309	Aircraft Armament Characteristics Reports Preparation of

WEAPON REQUIREMENTS

WR-8	Revised Pages for Aircraft Contract Detail Specifications
WR-11	Design and Test of Packaging, Packing, Shipping and Handling Equipment for Weapon System Components

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WR-29	Compatibility, Frequency Allocation, and Equipment Spectrum Signature: Requirements for
WR-43	Preparation of Quality Assurance Provisions
WR-62	Specification and Standards, Use of

(When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications, standards, drawings, and publications may be obtained upon applications to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120).

2.2 OTHER PUBLICATIONS. - The following documents form a part of this specification. Unless otherwise indicated in the applicable addendum to this specification, the issue in effect on date of invitation for bids shall apply.

2.2.1 STANDARDIZATION MANUALS.

M200	Standardization Policies, Procedures and Instructions

(Application for copies of the above manual should be addressed to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120).

2.2.2 CATALOGING HANDBOOK.

H-6	Federal Item Identification Guides for Supply Cataloging

(Application for copies of the above handbook should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402).

2.2.3 FORMS.

NAVWEPS 4200/25	Engineering Drawings and Associated Data Requirements
DD-816	Electron Tube Complement Report for Armed Services Equipment (Including Semi-conductor Devices)
DD-1423	Contract Data Requirements List
NAVAIR 13100/4	Descriptive Arrangement - Tankage Drawings

(Copies of the above forms should be obtained from the local Government representative).

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2.2.4 LISTS.

NAVAIR 16-I-525 Preferred Standard Test Equipment for Naval Aircraft and Guided Missile Electronics Equipment

(Application for copies of the above list should be addressed to Commanding Officer, Naval Air Technical Services Facility, 700 Robbins Avenue, Philadelphia, Pennsylvania 19111).

2.2.5 DOCUMENTS.

IRIG Document No. 102-61
IRIG Standard Coordinate System and Data Format for Antenna Patterns

(Application for copies of the above document should be addressed to: Secretariat, Range Commanders Council, White Sands Missile Range, New Mexico 88002).

INDUSTRY STANDARDIZATION DOCUMENT ASA Y32.16-1965
Electrical and Electronics Reference Designations

(Application for copies of the above document should be addressed to: American Standards Association, Inc., 10 East 40th St., New York, N.Y. 10016).

2.3 USE OF SPECIFICATIONS AND STANDARDS. - Use of specifications and standards by design activities shall be in accordance with WR-62. Approval of contractor documents for a specific contract shall not constitute approval for other contracts held by the same contractor or for contracts held by other contractors.

3. REQUIREMENTS.

3.1 ACTION. - Action required by NAVAIR, LGR's, etc., on engineering data and tests for target systems shall be as specified in paragraph 4.2 and Table I. NAVAIR acceptance or release action as indicated in Table I may be delegated to the Local Government Representative at the discretion of NAVAIR.

3.1.1 DATA TO BE FURNISHED. - No data is required by this specification or by applicable documents referenced in Section 2, unless specified in the contract or order. Data to be furnished shall consist of that listed on the completed DD Form 1423 referenced in, or made a part of the contract, unless otherwise specified in the contract.

3.2 QUANTITY, TYPE AND DISTRIBUTION.

3.2.1 QUANTITY. - Quantity of design data shall be as specified in Table I.

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3.2.2 TYPE. - Type of engineering data required shall be as specified in Table I. The data and drawings shall be furnished as follows:

(1) Non-drawing data, non-reproducible copies, shall be in accordance with MIL-D-5480, Type I, Class 1.

(2) Non-drawing data, reproducible copies, shall be in accordance with MIL-D-5480, Type I, Class 2.

(3) Drawings, non-reproducible copies, shall be in accordance with MIL-D-5480, Type II, Class 1.

(4) Drawings, reproducible, shall be as specified in 3.4.48 of this specification.

3.2.3 DISTRIBUTION. - Distribution shall be as specified in Table I.

3.3 REPORTS. - All reports shall be in accordance with MIL-R-18136.

3.4 ENGINEERING DATA REQUIRED. - The following engineering data are required.

3.4.1 PROGRAM MASTER PLAN. - A program master plan, designed to fulfill the basic programming requirements of NAVAIR and the contractor, shall be submitted. The plan shall be furnished in a ring binder suitable for the insertion of new and revised pages. The plan shall include:

(1) Brief description of the target system.

(2) A statement indicating the government-furnished equipment required including quantities, intended use and justification therefor.

(3) A statement indicating the government facilities required including intended use and justification therefor.

(4) A statement of program status in relation to schedules.

(5) Master schedule showing the following as applicable for the system and for each of the major components of the system (e.g., target, target control system, launcher, handling equipment, test equipment, packing, trainers and facilities):

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(a) R&D contract award
(b) Mockup approved
(c) Delivery of first developmental model
(d) First flight
(e) Initiation of contractor's demonstration
(f) Initiation of Navy technical evaluation
(g) Initiation of Navy operational evaluation
(h) Production contract award
(i) First production model delivered
(j) Fleet release

(6) A component development schedule showing the following as applicable for each item listed in the Master schedule with appropriate component breakdown (e.g., for the target; airframe, guidance, propulsion, etc.):

(a) R&D contract award
(b) Mockup approved
(c) Development model completed
(d) First flight or test of equipment
(e) Production prototype completed
(f) Design approval tests
(g) Production configured equipment available for Navy technical evaluation

(7) Detailed scheduled charts showing:

(a) When government-furnished equipments will be required
(b) The schedule for each of the development phases, test and study areas described in the program planning report of 3.4.2.
(c) Design, fabrication and delivery of each block of targets and supporting equipment
(d) Target dates in connection with each planned procurement for each of the following:
  (1) All specifications required for initiation of procurement
  (2) Request for Proposal from NAVAIR to contractor
  (3) Submission of proposal by contractor
  (4) Contract award in order to meet projected delivery schedule

(8) Allocation charts for equipment to be delivered

(9) Summary of existing contract items

(10) Graphical and tabular fiscal summary for each existing contract including projected expenditure rate

The master plan shall incorporate schedules covering a period of at least two years subsequent to the current fiscal year. The plan shall be updated or reissued quarterly to reflect the latest status of the program.

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The following statement shall be incorporated in a foreword or introduction in the report:

> "It is the purpose of this Master Plan to present, in one volume, a general picture of the scope of the program and sufficient schedule information to enable orderly planning of the future of the system by the contractor and NAVAIR personnel.
>
> Care should be exercised by persons using this Master Plan in that future quantities quoted are only estimates and as such do not necessarily reflect the actual requirements of the Government.
>
> Schedules are subject to revision without notice. Acceptance of this report does not imply concurrence in schedules and quantities by the Government nor should they be used in any way by the contractor to commit or obligate funds."

3.4.2 DEVELOPMENT PROGRAM PLANNING REPORT. - A development program planning report shall be submitted. The report shall be furnished in a ring binder suitable for the insertion of new and revised pages. The report shall include:

(1) A brief summary of the development program plan and a detailed description of the following phases of the program:

(a) Initial Design Phase which shall encompass the development of the basic target system design and associated data through Mockup.

(b) Prototype Development Phase which shall encompass the development from the Initial Design Phase through the construction of a functional prototype.

(c) Preflight Phase which shall encompass the development from the functional prototype through the construction of working models.

(d) Contractors Development Testing Phase which shall encompass the continued development from the working models to the contractor's demonstration.

(e) Contractor's Demonstration which shall include a brief description of the plans for, and facilities required for the contractor's demonstration. Duplication of the details of 3.4.45 is not required.

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(f) Evaluation Phase which shall encompass the continued development work from contractor's demonstration to Navy technical evaluation.

(g) Product Improvement Phase which shall encompass the development work to eliminate deficiencies and incorporate improvements considered necessary as a result of the technical and fleet evaluation.

(2) The description shall include but not necessarily be limited to the scope and general procedures to be followed in each of the following areas:

Development Tests
Reliability Tests (Contractor may refer to the report required by 3.4.3).
General Flight Tests
Captive Flight Tests
Target Launchings
Ground Handling Studies
Interagency Liaison
Safety Tests

(3) The planning report shall provide for program check points designed to promote orderly technical monitoring of the contractor's development program. (See 3.4.8)

3.4.3 RELIABILITY REPORTS. - The reliability reports shall be in accordance with MIL-STD-1304 and shall include the following:

(1) Reliability program plan

(2) Effectiveness study

(3) Reliability block diagram and mathematical model

(4) Reliability prediction

(5) Reliability allocation

(6) Criticality study

(7) Failure modes and effects study

(8) Environmental study

(9) Reliability test plan and test result

(10) Failure summary

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(11) Reliability status
- (a) Pert/Time management summary
- (b) Pert/Cost analysis
- (c) Reliability growth summary
- (d) Demonstrated reliability
- (e) Predicted reliability
- (f) Allocated reliability

3.4.4 ENVIRONMENTAL CRITERIA REPORT. - The report shall present the results of ground and flight tests performed to determine the fundamental environment in which the target must operate, and shall present the proposed methods for feeding this information into the design phase. Criteria shall be furnished with respect to the following:

(1) Shock and vibration caused by motor-burning, transportation, handling or flight.

(2) Altitude, temperature, pressure extremes.

(3) Launching airplane influences where applicable.

(4) Shipboard stowage, handling, and checkout restrictions as defined by NAVAIR.

The report shall present the design criteria for target system components as determined from the results of these tests and any other applicable information. It shall include:

- (a) Design, shock and vibration spectrum.
- (b) Design impact loading conditions.
- (c) Design internal temperature conditions.
- (d) Design steady state and gust load acceleration limits.

3.4.5 TARGET SYSTEM BROCHURE. - The brochure shall contain sufficient information to enable other interested organizations to become familiar with the general design and tactical features of the Target system for planning purposes. The brochure shall contain the following information:

(1) Introduction (General description of the Target, the Target system, and method of use.)

(2) Operational diagrams

(3) Weight and size of system equipment

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A preproduction preliminary copy of the document containing proposed art and manuscript shall be submitted for review and approval prior to preparation of final copy. Illustrations shall be used only as required for understanding of technical details. Final copy shall be submitted after incorporation of NAVAIR comments. Any and all classified and/or proprietary data contained in the document shall be clearly set forth on page 2 of the cover.

3.4.6 PROGRAM EVALUATION AND REVIEW TECHNIQUE (PERT) REPORTING REQUIREMENTS. - Pert/Time and Pert/Cost diagrams, estimates and reports shall be submitted as mutually determined between the contractor and the Navy. Diagrams, estimates and reports shall be as specified and in accordance with MIL-P-23189.

3.4.7 MONTHLY PROGRESS LETTERS. - These letters shall summarize the progress made in all phases of the design, all problems encountered during the reporting period, the causes of the problems and the proposed corrective action, including the time period required to make the correction, test, manufacturing and reliability programs during the period covered by the letter, and shall outline the work to be undertaken during the next period. It shall summarize important decisions and present questions for consideration. It shall contain a list of visitors to the project and visits made by the contractor's personnel to other agencies or contractors concerned.

3.4.8 DEVELOPMENT CHECK POINTS. - The following check points will be used to assist in evaluating and monitoring the development of the target system.

(1) INITIAL TARGET SYSTEM PROGRAM REVIEW. - This program review will take place approximately six months after contract date. The contractor shall be prepared to present detail information covering the work accomplished to date and discuss the design, engineering, scheduling and fiscal aspects of the items specified in 3.4.1 and 3.4.2. In addition, the contractor shall be prepared to discuss in detail all problem areas and other aspects of the program as requested by NAVAIR.

(2) MOCKUP. - When a mockup of the target or target system is required under the contract, the mockup shall be constructed for inspection by a Mockup Board as specified by NAVAIR. The mockup shall be constructed in accordance with MIL-M-18828, except that the mockup shall include the complete system. In constructing the mockup, emphasis shall be placed on overall physical arrangement and dimensions of the target and supporting equipment (test equipment, handling equipment, launch aircraft equipment and packaging). Its serviceability provisions and its aircraft and shipboard compatibility shall be illustrated. Mockup photographs shall be furnished in accordance with MIL-M-18828.

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(3) FUTURE TARGET SYSTEM PROGRAM REVIEWS. - Program reviews to the termination of the program will occur at approximately six month intervals and will be held alternately between NAVAIR and the contractor's plant. The contractor shall be prepared to present detail information covering the work accomplished to date and to discuss the design, engineering, scheduling and fiscal aspects of the items specified in 3.4.1 and 3.4.2. In addition, the contractor shall be prepared to discuss in detail all problem areas and other aspects of the program as requested by NAVAIR.

3.4.9 PRESENTATION DATA. - Presentation data shall include slides, charts, graphs, etc. and motion picture film footage.

3.4.9.1 SLIDES, CHARTS, GRAPHS, ETC. - The contractor shall provide slides, charts, graphs, etc. as required by NAVAIR for presentation purposes to portray up-to-date information on target system development and configuration. Slides shall consist of 3 1/4 inch by 4 inch lantern slides and 35 mm slides in color. An identifying description of the submitted material shall accompany this data.

3.4.9.2 MOTION PICTURE FILM. - Motion picture film shall be submitted in accordance with the following:

(1) COVERAGE. - The contractor shall accomplish motion picture photography on both a continuing and an "as specified" basis as indicated herein. Photographic coverage may include documentation of highlight aspects of research and development, facilities construction and utilization, hardware fabrication and production, test activities, including high speed engineering sequential motion picture coverage of appropriate test phases, and related events and subjects involving the contractor's area of responsibility.

(2) OBJECTIVE. - The objective of the motion picture coverage specified herein is to satisfy a continuing need for documentation and reporting of the contractors research and development activities and progress. The motion picture coverage thus obtained will be used for purposes of program evaluation and management, analysis, written report backup, and the preparation of training, orientation and briefing films. Other uses include legal, historical, and the fulfillment of various information services requirements.

(3) GENERAL. -

(a) All motion picture coverage will be professional photographic quality and in a quantity that will enable the Navy and the contractor to produce factual, minimum-cost film reports, and the Navy to produce the film clips indicated in subsequent paragraphs.

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(b) Original motion picture photography will be in 16 mm color, exposed at 24 frames per second. Black-and-white film and other frame-rates and other film sizes may be used in instances where the capability of the color film or the normal frame-rates would be detrimental to the accomplishment of specialized photographic coverage, such as aerial, engineering sequential, and time measurement photography.

(c) The camera film shall be that type of raw stock which will best suit the conditions surrounding the particular coverage assignment.

(d) The contractor shall not project or cut original film exposed in connection with the contract, except to eliminate waste film caused by camera failure or faulty photographic techniques (gross over or underexposure, over or under development, out of focus,etc.) which results in qualitatively unsatisfactory film.

(e) Completed film reports shall contain no contractor promotional material beyond the name of the firm in the title and in such places in the picture and narration where it is needed for purposes of identification and clarity.

(f) All original camera film footage shall be slated whenever possible. Slate information shall include, as appropriate: contractor identification, project number and/or name, contract number, security classification, date photographed, scene and take number. One copy of caption information describing the action involved in each scene and the significance of the sequence of which the individual scene is a part, will be forwarded with all original camera film footage. All individual reels of the film footage will bear head and tail security classification leaders.

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(g) All of the original motion picture footage produced and costed under terms of this contract shall be the property of the Commander, Naval Air Systems Command. Unless otherwise specified herein, the contractor shall forward all original motion picture material produced under this contract within 120 days after processing. Exception will be made on such portions of the original footage as may be required for the production of planned complete film reports called for under terms of this contract. In this latter circumstance the motion picture original material will be forwarded not later than one year after processing.

(h) All classified motion picture film footage, and completed film reports produced under this contract will be handled in accordance with the security regulations.

(4) PRODUCTION AND SUBMITTAL OF MOTION PICTURE DOCUMENTATION MATERIALS:

(a) The contractor shall accomplish continuing motion picture coverage of all significant high light events within the area of his activity and responsibility as they occur and as he determines is essential to the fulfillment of NAVAIR's need for engineering, evaluation, and management data, or for reporting purposes as indicated in subsequent section of this specification. Coverage will include the unsuccessful and unfavorable events as well as the positive aspects of the contractor's activity and progress. The contractor shall also accomplish coverage specific developments and special events within the area of his activity and responsibility, as may be directed. Such special coverage may coincide with the continuing coverage described above or it may be an additional requirement.

(b) The contractor shall furnish the following completed motion picture reports, and selected film footage for production of film clips, from the photography accomplished in paragraph (4)(a) above:

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(1) Progress Film Reports:

(a) The contractor shall produce formal progress film reports annually.

(b) Progress film reports shall be complete audio-visual reports organized objectively to give a factual account of a phase or portion of the contractor's project or program, its development, production progress and test activity. These film reports shall be produced in a manner which will permit compilation of an integrated and chronological film record of the entire history of the progress and activity within the contractor's area of work and responsibility when all progress film reports from the contractor are combined.

(c) Progress film reports shall be approximately 5 to 20 minutes in length (1/2 to 2 reels), depending upon the reporting period, and shall normally contain only picture, titles and narration. However, when it is deemed essential for clarity, a limited amount of simple animation, artwork, graphs, lip sync, and sound effects may be used.

(d) Title format specifications are reflected in Figures 1 and 2 and shall be used for all film reports produced under terms of this contract.

(e) The contractor shall prepare and deliver the items listed below on, or prior to the dates specified:

(1) Release prints shall be delivered for information, not later than 40 calendar days following the end of the reporting period.

(2) All pre-print materials used to make the release prints in (4)(e)(1) above, including the matching sound track negative, and one copy of the final cutting and recording continuity script, shall be delivered not later than 60 calendar days following shipment of the release prints. Prior to shipment of pre-print materials, the contractor may produce one release print for his own use.

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(2) Special Film Reports. - The production of special film reports may be assigned to the contractor from time to time by the contracting officer. Such films will encompass scientific, engineering, and technical aspects of the research, development, fabrication, and testing phases of the contractor's area of activity and responsibility. Special film reports shall conform to the same requirements specified above for progress film reports, and when directed by the office of primary responsibility, prints will be made and distributed to the same addressee. The production of these special film reports will be specifically directed by the contracting officer and will be negotiated individually as required.

(3) Film Clip Footage:

(a) In addition to the completed film reports called for in (4)(b)(1) and (4)(b)(2), the contractor shall submit on an expedited basis film footage selected from the record and documentary motion picture photography accomplished in compliance with (4)(a) above, or obtain and submit such film footage specifically filmed for the preparation of film clips as directed. Subject matter of this film footage will include coverage of special happenings such as program reviews, mock-ups, test flights, critical or complex problem areas, technological break-throughs, engineering sequential coverage of research, development and laboratory test activities, and other events which depict the program progress and status. Such footage will be used by the Navy to prepare clips for the in-service production of films and for other reporting and briefing purposes.

(b) Footage for film clips suitable for use in the various film reports must portray a complete story of a specific research and development event, phase, or activity. The photographic coverage for each clip should include a variety of scenes of the reported item or event, i.e., establishing shots, medium shots, close-ups, and cutaways, to assist the Navy film editor in telling the story. The film footage should consist of full length, unedited, untitled, silent

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scenes assembled in continuity and of sufficient length to provide 5-10 minutes running time as received from the contractor.

(c) One 16 mm, color, timed master bearing emulsion in the original camera position, and one matching color work print of the film clip footage shall be included. Both the master and the work print shall be edge numbered or coded. Two copies of written information describing the activities and items shown shall be delivered with this footage.

(d) The completed motion picture reports and film clips footage shall be limited to five prints for any single report and shall be coordinated through the Chief of Information with the appropriate Department of Defense Office if the report is to be released to the public or if it deals with research and development projects.

(5) REPORTING OF MOTION PICTURE COVERAGE. - The contractor shall forward a written report on each completed film report and each shipment of motion picture footage submitted for the preparation of film clips. The following information shall be provided in narrative form or in the format shown in Figure 3:

Subject and/or title of the photography (completed film report or film clip footage), classification, size and type of film, contract number, project name and number, purposes of photography (e.g., quarterly progress report, film clip, OPR request, etc.), location of original film. (NOTE: If original film has been forwarded, so indicate and give date of shipment; if not submitted give estimated date of shipment, number and location of masters and other pre-print materials (if in the possession of contractor, give estimated date of shipment, number and location of prints) (for completed film reports indicate number and location of composite release prints), estimated cost (original photography should be costed to the completed progress or special film reports when filmed for inclusion in such reports; original photography specifically filmed for film clip purposes should be costed to the particular film clip involved)).

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(6) PROCEDURE AT NAVAL FACILITIES. - When any portion of the work conducted under this contract is performed at a Naval Test Range or other Naval Facility where a technical motion picture production capability exists, the Naval Facility will accomplish the photography and provide the contractor with the original or a 16 mm color, timed printing master with emulsion in the camera position and one matching color work print. It will be the responsibility of the contractor to provide the liaison and technical guidance to assure that all applicable portions of the contractor's program are covered.

3.4.10 SIMULATOR PROGRAM REPORTS. - A report shall be prepared for each major simulator program. The report shall state the objectives of the program, the duration of the program, the simulation set-up employed, the results of the study, the degree to which the objectives were fulfilled, conclusions, and recommended action.

3.4.11 AIRCRAFT INSTALLATIONS REPORT. - An aircraft installations report shall be submitted. The report shall contain sufficient information to permit incorporation of the target system in the launching aircraft by the aircraft contractor or by the target contractor. The report shall also include the following:

(1) An analysis of the effect of the target system on the performance including maneuverability, speed and altitude of the launching aircraft listed in applicable target system detail specification, the characteristics of which will be furnished by NAVAIR.

(2) A description of all launching aircraft equipment, the weight, space, power, heating and cooling requirements, adapters and special fittings required for the aircraft installation. The arrangement of the equipment as installed or proposed for each of the aircraft listed in the applicable target system detail specification shall be shown.

(3) Clearly show deck and catapult bridle clearances on all carrier aircraft installations carrying targets externally.

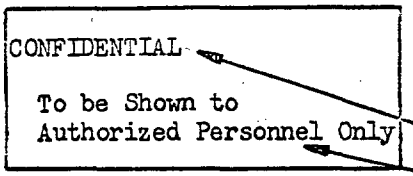
3.4.12 SHIP INSTALLATION REPORT. - A report shall be submitted containing sufficient information to permit incorporation of the target system in aircraft carriers specified by NAVAIR. The report shall include:

(1) Recommended shipboard allowances of all target components (including major components, wings, fins, connections, squibs, etc.) target and support equipment spare parts, handling and loading equipment, special adapters and fittings, tools (special and ordinary) for target and support equipment, shipping containers, target checkout equipment, test equipment for the target checkout equipment and any other item that must go on board ship to support the target system.

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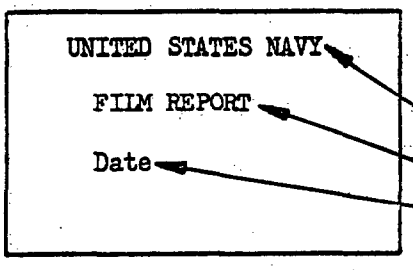
MOTION PICTURE TITLE FORMAT

SPECIFICATIONS



Title Classification (Beginning and End)

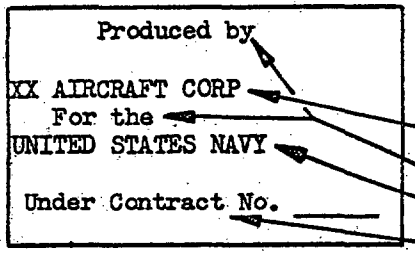
1. No. Drop Shadows
2. ACME - 10 Field
3. 120 PT. Franklin Gothic Extra Condensed
4. 36 PT. Bernhard Gothic Heavy



Identification Title

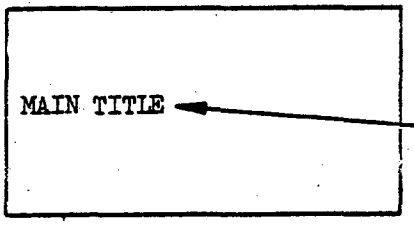
1. All lettering with Black Drop Shadows
2. ACME - 10 Field
3. 48 PT. Bernhard Gothic Medium Condensed
4. 84 PT. Franklin Gothic Extra Condensed
5. 36 PT. Bernhard Gothic Medium Condensed

Indicate period covered by report.



Credit Title

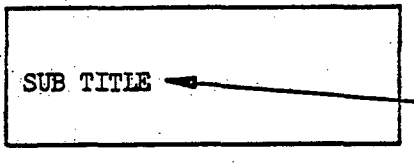
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2. ACME - 10 Field
3. Type face and size at discretion of Contractor
4. 48 PT. Kauffman
5. 48 PT. Bernhard Gothic Medium Condensed
6. 36 PT. Bernhard Gothic Medium Condensed



Main Title

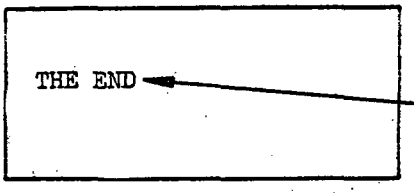
1. All lettering with Black Drop Shadows
2. ACME - 10 Field
3. 96 PT. Lydian Bold
4. Use unclassified title if possible

NOTE: Main titles should indicate subject matter contained in report.



Sub Title

1. All lettering with Black Drop Shadows
2. ACME - 10 Field
3. 84 PT. Lydian Bold
4. Use unclassified title if possible



End Title

1. No Drop Shadows
2. ACME - 10 Field
3. 96 PT. Lydian Bold

LETTERING: White
BACKGROUND: Navy Blue, textured

Figure 1

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PRODUCTION GUIDE FOR TITLING
(For Completed Films Only - Not Applicable to Film Clips)

LEGEND: Fl-Fade In FO-Fade Out FR-Frames

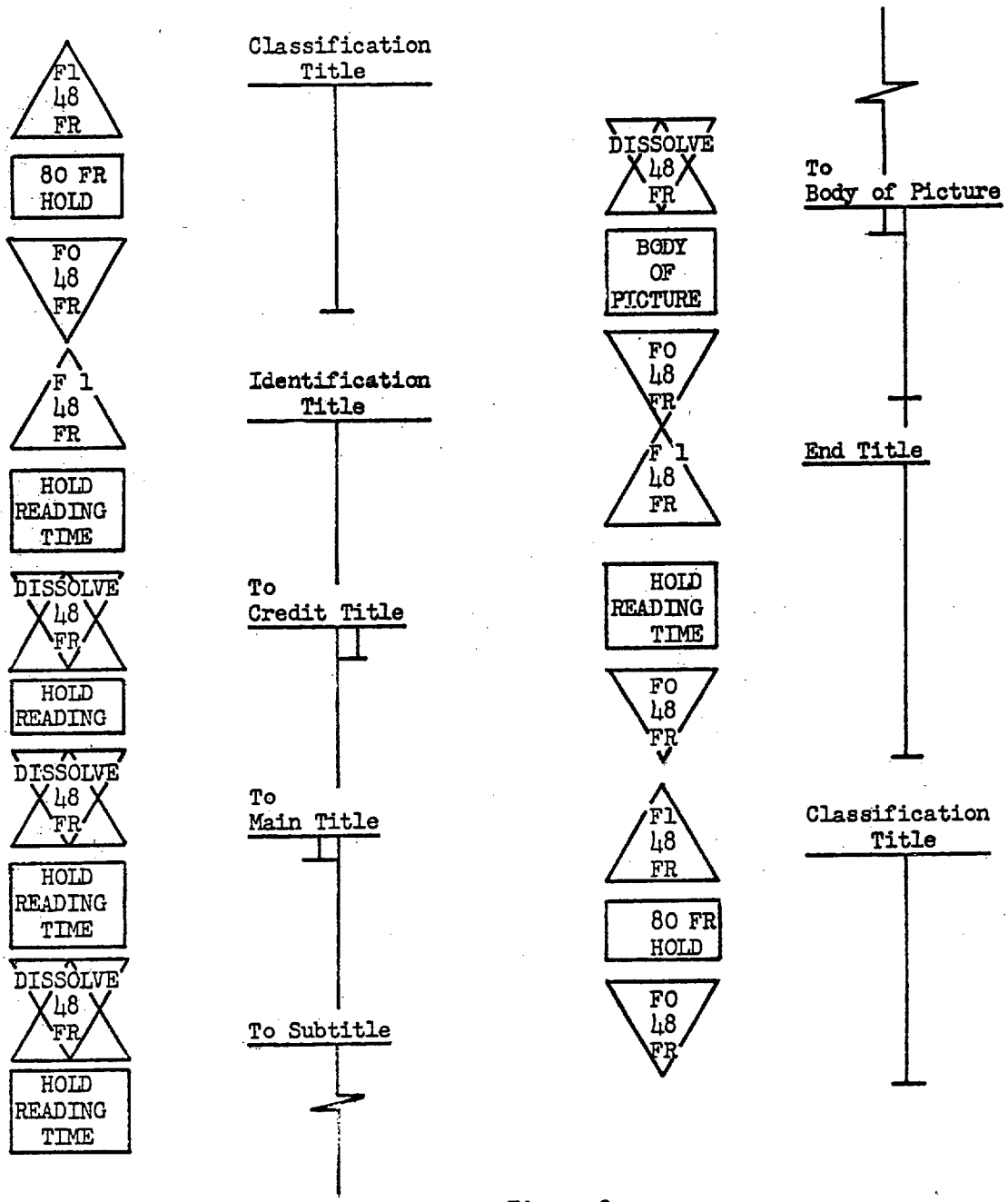


Figure 2

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REPORT OF PICTORIAL COVERAGE STILL AND/OR MOTION PICTURE PHOTOGRAPHY (To be prepared by contractor)				Period Covered by Report			Date of Report		Fiscal Number	
Item	Subject of Photography	Classification	Size Film	Type Film	Amount Exposed	Date Exposed	Photographer	Cont. Number	W/s or Proj Number	Estimated Cost
1										
2										
3										
4										
5										
	Purpose of Photography	Location (Where Exposed)		Location of Original Film	Number of Masters		Location of Masters		Number of Prints	Location of Prints
1										
2										
3										
4										
5										

Figure 3

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3.4.12 (Cont)

(2) Written descriptions and outline drawings of each item listed in paragraph 3.4.12(1), including for each item as applicable the weight, dimensions, center of gravity, hard spots for fastening, sensitive areas subject to damage, and any other pertinent details. In addition, foundation and mounting information, utilities required (air, nitrogen, electric power, etc.), heat generation data and wiring and piping connection details shall be furnished for all test equipment to be installed on ships.

(3) Estimated target assembly time, required shipboard readiness and recommended shipboard flow pattern. Description of flow patterns shall include flow from replenishment station through strike-down routes, stowage in magazines, breakout, check-out, strike-up, assembly/disassembly as applicable during these evolutions, aircraft de-arming following captive flight or after recovery operations as required. Flow diagrams shall show target configuration whether packaged or bare, handling equipment at each point in the flow pattern, along with necessary universal and/or special adapters. Flow patterns will also stipulate number of personnel required for each operation and estimated time required to complete the operation.

(4) Recommended shipboard stowage, testing and handling procedure. This phase of the report should set forth recommendations on locations of stowage and testing facilities; arrangements of all compartments; stowage requirements (bare or packaged); and shock mitigation, temperature and humidity requirements. Recommendations on handling methods should be the result of analyses of sizes and weights of components and complexity of assembly.

3.4.13 HANDLING EQUIPMENT REPORT. - This report shall describe all handling equipment required for use with the target system including standard items and special adapters thereto.

3.4.14 SHIPPING CONTAINERS REPORT. - A report shall be submitted describing the shipping containers for the target or target system as applicable. The report shall show provisions and design limitations for shock mounting, vibration isolation, sealing, lifting, loading on transportation medium, stacking and re-use in compliance with the criteria of WR-11 and MIL-S-901.

3.4.15 AERODYNAMIC AND PERFORMANCE DATA.

3.4.15.1 AERODYNAMIC INVESTIGATIONS.

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3.4.15.1.1 AERODYNAMIC INVESTIGATION PROGRAM REPORT. - A report shall be submitted showing the planned aerodynamic investigation program and schedule for development of the target design. This report shall outline the purpose and scope of each proposed investigation, indicate the test facilities to be employed, test dates desired, estimated occupancy time required, describe the scale and type of models to be constructed and tested, and present the ranges of test variables to be investigated. Program revisions shall be submitted for acceptance prior to undertaking the revised investigations. For wind-tunnel flutter model investigations, MIL-A-8870 shall apply.

(1) For investigations at Government-furnished facilities - Preliminary contact with the facility to be utilized is usually necessary to determine the feasibility and desirability of conducting the proposed investigation. Following such contact, a request for tests shall be made to NAVAIR. The request shall describe in detail the specific test programs agreed upon. For tests to be scheduled at the David Taylor Model Basin, submittal of requests shall be in accordance with published procedures.

(2) For investigations at privately operated facilities - A detailed description of tests to be conducted and of specific objectives of the tests shall be submitted to NAVAIR.

3.4.15.1.2 AERODYNAMIC MODEL DRAWINGS. - When an aerodynamic model or models are required under the contract, the design shall be suitable for carrying out the approved aerodynamic investigation program. In establishing model scale ratios, strength, and installations provisions, consideration shall be given to utilizing the maximum number of alternate aerodynamic test facilities. Prior to undertaking construction of any model, approval of the model design by the appropriate test facilities is required. Approved drawings, and design data required for construction of the models shall be submitted to the constructing facility as soon as practicable. For dynamic and free-flight model investigations, the contractor shall furnish the constructing and testing facilities with all design information necessary to simulate full-scale conditions in the construction and during the testing of the model. All model drawings shall be made available to the government on request.

3.4.15.1.3 AERODYNAMIC INVESTIGATION DATA. - Data from aerodynamic investigations shall be kept current by prompt delivery of aerodynamic reports in accordance with the following:

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(1) Interim letter reports. - These shall be submitted immediately following completion of each facility occupancy and shall cover items of interest such as purpose, scope, contractor's observations, difficulties encountered, significant results, and conclusions and recommendations as may be possible based on inspection of the available preliminary test results.

(2) Test Data reports. - Reports presenting the basic aerodynamic data and test results obtained from investigations conducted in contractor-furnished and private test facilities shall be submitted. Such reports should include description of the configurations tested and comparison with previously listed versions and with the current proposed target configurations. Test data should be shown graphically and be referred to axes consistent with the stability and control reports of 3.4.15.3 when possible. Tabulated data are not desired unless specifically requested. When subsequent analysis indicates any of the published data to be erroneous, the NAVAIR shall be informed and supplied with appropriate errata or replacement pages.

3.4.15.2 AERODYNAMIC AND THERMODYNAMIC HEATING DATA. - The following reports are required whenever the recovery temperature or local temperatures due to environment, operation of electronic equipment, etc. equals or exceeds 200°F.

3.4.15.2.1 AERODYNAMIC AND THERMODYNAMIC HEATING ANALYSIS REPORT. - This report shall present the contractor's analysis of the aerodynamic heating to which the target will be subjected and shall include a discussion of the method of analysis including explanation and justification of the assumptions and approximations used together with adequate references for all sources of information. A short discussion of the predicted accuracy of the analysis results is to be given. An explanation and justification for the selection of the target trajectories and launch conditions for the target system, is to be included. For each selected trajectory time history data for surface temperatures and compartmental temperatures shall be given. These data shall include the effects of environment prior to launching and any sources of heating or cooling within the target.

3.4.15.2.2 STRUCTURAL HEATING SUMMARY REPORT. - This report shall include data on the deformations due to differential heating during transients and deformations under stabilized temperature conditions, effect of deformation on airloads, and effect of temperature and deformation of stiffness and other parameters pertinent to flutter analysis. Additional data is to be given on the internal stresses due to differential heating and pertinent allowable stress data for structural design. From the calculated temperatures of local area (including the external surface, backup structure, and structure near power plants (both external or internal), electronics, or other heat sources and structure located in or near the engine exhaust stream), define or clearly reference the material properties

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used for design. These data shall include static properties at temperature and following exposure to temperature, consideration of loading rates, criticalness of creep parameters, effect of temperature or fatigue strength (when pertinent), joint relaxation data, and fastener, welding, etc. allowables.

3.4.15.2.3 COMPONENT HEATING SUMMARY REPORT. - This report shall present the contractor's analysis of the effects of aerodynamic and thermodynamic heating and the resultant structural deformations on the functioning of the target components. Compartmental and surface temperatures consistent with the data of the Aerodynamic and Thermodynamic Heating Analysis Report shall be used to analyze the suitability of operational environment upon the function of all internally stowed items such as fuel, engine, electronic gear, hydraulic systems, etc. Wherever heat protection of an item is required, the analysis shall be extended to substantiate the adequacy of the heat protection provisions.

3.4.15.3 STABILITY AND CONTROL REPORT. - This report shall contain values of the airframe static and dynamic stability and control derivatives (with and without booster) over the complete design envelope. Experimental data and similar configurations and results of wind tunnel tests, aeroelastic analysis, and thermoelastic analysis accomplished shall be included. This report shall also include the following data: (1) target configuration including three-view drawing, dimensions, and areas, (2) a list of symbols used including definitions and units, and (3) substantiating data including method used to obtain the aerodynamic derivatives. The information presented in this report shall be based upon the latest and most applicable experimental and theoretical data available at the time of submittal, and shall be fully documented and referenced. When major changes are made in the target, revisions to this report showing the effects of these changes shall be submitted as soon as possible and without regard to the normal date for periodic revision. At least 30 days prior to first flight, the data in this report shall be revised as necessary and the complete report submitted for acceptance prior to release of the first flight. In the event flight test data show different stability and control characteristics than predicted, an appendix shall be submitted to show these differences.

3.4.15.4 SYSTEM DYNAMICS REPORT. - This report shall include analyses of all airframe and stabilization system configurations and modes of operation. A sufficient number of flight conditions shall be analyzed so that a satisfactory degree of stability may be expected over the entire operating envelope. If an analogue computer is used in the analyses, substantiating "paper and pencil" analyses shall also be made for the selected values of the system parameters. This report should also include the following data: (1) A list of symbols used including definitions and units, (2) Tables of aerodynamic stability derivatives for the conditions analyzed, (3) Numerical coefficients of the airframe equations of motion for all conditions analyzed, (4) All system and subsystem transfer functions used in the analyses, (5) Numerical values of roots determined from computer systems or root locus analyses.

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3.4.15.5 CHARACTERISTICS AND PERFORMANCE DATA. - Performance data reports and Standard Aircraft Characteristics(SAC) charts and drawings prepared in accordance with the following subparagraphs and with Specification MIL-P-21018 are required as follows:

3.4.15.5.1 BASIC DATA. - Submit basic data prepared in accordance with MIL-P-21018 for acceptance prior to proceeding with performance calculations for submittal with SAC charts.

3.4.15.5.2 PERFORMANCE DATA REPORT.

(1) Performance Data Report. - Using basic data accepted or designated by NAVAIR, prepare and submit an initial Performance Data Report. Revised data shall be submitted whenever changes in target configuration or powerplant result in significant performance changes. The required format is outlined in paragraph 3.1.2 of Specification MIL-P-21018. The performance data report shall substantiate the performance data presented in the Handbook of Flight Operating Instructions.

(2) SAC Charts. - SAC charts shall be 10 7/8 X 14 1/4 inches centered on 12" X 15" matte prints in accordance with MIL-P-21018.

(a) Descriptive arrangement and armament and tankage drawings shall be submitted in the form of reproducible glossy prints prepared on NAVAIR Form 13100/4.

3.4.15.5.3 ADDITIONAL. - Additional submittals of revised Performance Data of 3.4.15.5.2 when required.

3.4.15.6 LAUNCHING CHARACTERISTICS DATA. - The data and reports requested below shall be provided unless the contractor is specifically exempted from launching characteristics responsibility. If so exempted the contractor shall provide such technical information as requested by NAVAIR to allow launching characteristics computations as indicated below to be performed by another agency.

3.4.15.6.1 LAUNCHING CHARACTERISTICS REPORT. - The contractor shall prepare and submit a report or reports analyzing the separation characteristics and launch trajectories of the target(s) under the most critical launching conditions.

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(1) AIR LAUNCHED TARGETS. - The analyses shall show the time histories of the motion of the target(s) from initial firing or ejection to stabilized flight free of the flow field induced by the launching aircraft. The analysis shall be made for all target locations on all launching aircraft assigned for conditions deemed critical and shall include justification of the conditions selected. The analysis method shall be briefly presented with appropriate references and explanation of choice. Launching aircraft induced flow field effects and the calculated transient effect of target firing upon engine operations are to be included. Possible compressor stall and flameout shall be considered as critical conditions in the analysis. An analysis of jettison trajectories to insure that the target(s) safely clear(s) all aircraft structural components will be made where applicable. Available information on the launching aircraft will be requested from the Government.

(2) SURFACE LAUNCHED TARGETS. - The analysis shall include a thorough investigation of the effects of ship motion, including pitch, roll and heave, and the effect of wind on the trajectory of the target. Included will be the effects of the highest sea state in which the target is capable of being launched and in addition the effects of the wind from all applicable directions, or any other natural phenomenon which may limit the launch envelope. The loads on the launcher resulting from the above ship motions and wind will be calculated for all conditions. The path and/or effect of any boost assist will be included in the analysis. In cases where applicable, mention will be made as to the amount of sea water to which the target will be exposed.

3.4.16 STRUCTURAL DESIGN, ANALYSES, AND TEST DATA. - Structural design, analyses, and test data shall be submitted for acceptance in accordance with MIL-A-8868 and MIL-A-8870 covering structural, flutter and divergence, and vibration requirements respectively. However, in lieu of the delivery schedules specified in MIL-A-8868 for stress analysis reports and loads reports, such reports shall be submitted not later than 30 days prior to first flight or not later than 60 days prior to commencing static tests, whichever occurs first.

3.4.16.1 DRAWINGS. - Drawings of major structural components shall be submitted showing structural design features including important fitting attachments, beams, ribs, bulkheads, method of skin reinforcement, carry-through structures, joints, splices, cutouts, and other such discontinuities, typical sections through load-carrying members, materials employed and heat treatment, types of riveting, types of welding, and other methods of attachment of important parts. Submittal of drawings shall not be delayed until complete assembly, installation, or production drawings are available. All authorized deviations from detail or other applicable design specification requirements shall be indicated to facilitate review of the drawings. These deviations shall be marked on the drawings in colored crayon, the color of which shall be in sharp contrast with the color of the print.

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Release of drawings under this item shall not constitute official release of authorized deviations from detail or design specification requirements and shall not be construed as acceptance of non-government specifications or standards. Drawings of major structural components are required as follows:

(1) General Arrangement. - (front and side elevations and plan views)
(2) Wing Group
(3) Control and Stabilizing Surfaces Group
(4) Body Group

3.4.17 ELECTRIC SYSTEMS DATA. - The following data shall be submitted. Equipment shall be identified by Military or manufacturer's part number. Symbols shall be in accordance with MIL-STD-15 and reference designations in accordance with ASA Y32.16-1965 Document. Data shall include information on the complete target system or target as applicable.

3.4.17.1 INFORMATION ON NON-STANDARD PARTS. - Requests for approval of non-standard parts shall be accompanied by the data required by MIL-E-8189 or MIL-E-5400 as applicable.

3.4.17.2 NOMENCLATURE, NAMEPLATES AND SERIAL NUMBERS. - Nomenclature, nameplates and serial numbers shall be in accordance with MIL-N-18307. The contractor shall request nomenclature for the articles, equipments and major components thereof giving the information required by the Federal Item Identification Guides for Supply Cataloging (Cataloging Handbook H6). Two copies of dimensional outline drawings of each article, equipment or major component thereof shall accompany the nomenclature requests.

3.4.17.3 SCHEMATIC DIAGRAMS. - Preliminary schematic system diagrams in elementary form showing the complete electrical system in straight line, shall be supplied. Sufficient information shall be provided to facilitate engineering review and to clearly show the functioning of each electrical circuit.

3.4.17.4 MASTER WIRING DIAGRAMS. - Master wiring diagrams plus individual system schematics shall be supplied. These diagrams shall be suitable for incorporation in the applicable target or target system handbook. Each system shall be presented on one sheet, if practicable, without undue crowding. Complete drawings shall be provided for circuit elements, such as terminal blocks and connectors which are common to more than one system. Terminal blocks, connectors and wiring shall be identified to correlate with the physical installation.

3.4.17.5 AIRCRAFT-TARGET ELECTRIC SYSTEM DATA. - When the target system includes an air-launched target, the master wiring diagrams shall include the following:

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(1) The tie-in of the target system to the launching aircraft generating equipment showing both AC and DC information, i.e., frequency, voltage requirements, current requirements and power factor. Power from the launching aircraft will conform to MIL-STD-704.

(2) The use of power conversion devices required in the launching aircraft, e.g. rectifiers, inverters.

(3) The use of electric and electronic equipment required in the launching aircraft that is peculiar to the system, e.g. firing panels, illumination, controls.

(4) The type(s) of umbilical plugs and receptacles used both in the aircraft and the target.

(5) The information that is transmitted to the target on each of the umbilical pin numbers, including information as to voltage and approximate current required by the target through these pins.

(6) The drawing(s) shall be in sufficient detail to follow the step-by-step launching sequence of events both in the target and in the aircraft. The target circuitry may be represented in equivalent form; the launching aircraft components may be represented in semi-schematic block diagram form as long as the launching sequence can be traced.

3.4.17.6 ELECTRICAL LOAD ANALYSIS. - Preliminary and final electrical load analysis shall be submitted for the following as applicable.

3.4.17.6.1 LAUNCHING AIRCRAFT. - An electrical load analysis of the launching aircraft as modified to accommodate the target system in accordance with MIL-E-7016.

3.4.17.6.2 TARGET. - An electrical load analysis of the target during the post launch period until impact or recovery as applicable in general accordance with MIL-E-7016.

3.4.18 MATERIALS AND PROCESSES DEVELOPMENT AND EVALUATION REPORT. - A summary technical report describing materials and processes research, development and evaluation work which has been conducted or is planned under the contract, or alternatively, a statement that no work of this type has been conducted or is contemplated shall be submitted.

3.4.19 MAGNESIUM ALLOY PARTS LIST. - A list of all magnesium alloy applications in the target, together with the alloys used and applications if not obvious from the titles, a description of the parts and their applications shall be included.

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3.4.20 FINISH SPECIFICATION. - A finish specification in accordance with MIL-F-7179 shall be furnished. The contractor shall number paragraphs to correspond with the system employed in Specification MIL-F-7179 and shall indicate in the submitted specification and the forwarding letter all desired deviations from MIL-F-7179 and other applicable specifications. The reasons for deviations and supporting technical information shall be submitted with the finish specification to NAVAIR for approval.

3.4.21 HEATING AND COOLING REPORT. - A report shall be submitted describing the heating, cooling, insulation, anti-icing and defrosting provisions as applicable.

3.4.22 HYDRAULIC SYSTEM DATA. - Hydraulic system data shall be submitted as required by MIL-H-25475.

3.4.23 PNEUMATIC SYSTEM DATA. - Pneumatic system data shall be submitted as required by MIL-P-5518.

3.4.24 RECOVERY SYSTEM REPORT. - A report shall be submitted describing the target recovery system in accordance with MIL-P-25062. The report shall include drawings of the attachment provisions, release mechanisms, and stowage provisions.

3.4.25 HUMAN FACTORS DATA. - Reports shall be submitted as required by MIL-H-22174 for use in monitoring and evaluating the contractor's human factors effort in the design and development of the target system.

3.4.26 ELECTRONIC EQUIPMENT DATA. - For each electronic equipment developed by the contractor for the target system, the following data shall be submitted. Electronic equipment shall include flight test and maintenance and operational test and checkout equipment.

(1) INFORMATION ON NON-STANDARD PARTS. - Requests for approval of nonstandard parts shall be accompanied by the data required by MIL-E-8189 or MIL-E-5400 as applicable.

(2) ELECTRONIC TUBE COMPLEMENT REPORT. - For each equipment which employs electronic tubes (including crystal diodes and transistors) a report containing the proposed tube complement shall be submitted. The report shall be submitted on DD Form 816 which may be obtained from the Office of Printing and Publication of the various Naval Districts.

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(3) NOMENCLATURE, NAMEPLATES AND SERIAL NUMBERS. - Nomenclature, name-plates and serial numbers shall be in accordance with MIL-N-18307. The contractor shall request nomenclature for the articles, equipments and major components thereof giving the information required by the Federal Item Identification Guides for Supply Cataloging (Cataloging Handbook H6). Two copies of dimensional outline drawings of each article, equipment or major component thereof shall accompany the nomenclature requests.

(4) DATA TO BE SUPPLIED WITH PRE-PRODUCTION OR DESIGN APPROVAL MODEL. - When a pre-production or design approval model of equipment is required under a target system contract, two copies of each item of the following data shall accompany the model when shipped for test:

(a) External wiring diagrams

(b) Practical wiring diagram of each component or each constructional unit thereof, whichever is practicable, showing physical location and connections of detail parts and subassemblies with reference symbols and terminal numbers indicated.

(c) Complete schematic diagram

(d) Overall functional block diagram

(e) Brief operating instructions

(f) Outline dimensional sketches of all major components

(g) Report of contractor's tests

3.4.27 CONTROL AND STABILIZATION SYSTEM REPORT. - A system design report shall be submitted accurately describing and analyzing the performance, stability and operating characteristics (dynamically and statically) of all mechanical, electro-mechanical, pneumatic, hydraulic control and stabilization equipment for each of the following portions of the flight path for each configuration: (1) Launching; (2) Mid-course; (3) Terminal or Recovery as applicable. Power requirements, power developed by the system, type of input, amplification system, system parameters, and response characteristics shall also be included.

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This report shall also include a detail description of the methods and tests employed in making the analysis. This report shall be furnished as it evolves throughout the development phases and shall consist of an up-to-date description of the complete control and stabilization system. A revised analysis shall be submitted prior to flight test in all cases of change of either stabilization system or target characteristics which may reasonably be expected to affect flight performance. Control and stabilization system components used in these tests shall be available to the Government at any time after the tests are completed for additional or special tests as may be required. Functional block diagrams and schematics of all equipments and mechanisms shall be presented.

3.4.28 GUIDANCE SYSTEM REPORT. - A report shall be submitted which sets forth the guidance problem and describes in detail the contractor's design approach to the solution of the problem. This report shall include a complete frequency analysis, system drawings such as block diagrams with functions explained and schematic diagrams, including antennas with functions of each element explained. These data shall be furnished as they evolve throughout the development phase and shall consist of an up-to-date description of the complete guidance system, including the drawings and functional explanations specified above.

3.4.29 MAINTENANCE AND OPERATIONAL TEST AND CHECKOUT EQUIPMENT REPORT. - A report shall be submitted describing the procedures for testing the target system from acceptance to operational use. The report shall include all testing to be performed by operating forces and in service shops ashore and afloat. Insofar as practicable, items of test equipment shall be selected from the list of preferred standard test equipment, NAVAIR 16-1-525. The following information shall be furnished by the contractor for each item of recommended special test equipment:

(1) Identification of the equipment for which the test item is being recommended

(2) Identification of the recommended special test item.

(3) A brief technical description of the anticipated use of recommended items e.g. neasure magnetron power; display band pass audio filters; rotate gyroscopes at an accurately adjustable rate.

(4) A brief technical description of all required electrical and mechanical characteristics. This description shall be in sufficient detail to allow NAVAIR to determine if the functions of the special test equipment can be performed by alternate items.

3.4.30 FLIGHT TEST EQUIPMENT REPORT. - A report shall be submitted describing the flight test equipment, including telemetering equipment, beacons, etc., required during the flight test program.

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3.4.31 TELEMETERING ELECTRONIC DESIGN DATA. - Provide data for Telemetry Sets for the individual requirements of MIL-D-18300 as defined below. (Numbers below refer to Table I of MIL-D-18300).

3.5 Test Procedures for Individual Acceptance Tests Only. - Copies of these procedures shall be furnished the LGR for use in Acceptance Testing.

3.7 Report on Contractor Tests

3.8 Nomenclature; Serial Numbers; and Pertinent Identification Data

3.11 Prepare drafts of any proposed revisions to existing specifications or new specifications, required to reflect the design of the equipment being produced.

3.18 Frequency Allocation and Spectrum Signature Data

3.4.32 ANTENNA DATA REPORT. - An antenna report shall be submitted which shall show the proposed general arrangement of all antennas. In addition, the report shall include the following data for each antenna:

(1) Radiation patterns of the antennas installed on the target vehicle in accordance with IRIG Document No. 102-61, IRIG Standard Coordinate System and Data Format for Antenna Patterns.

(2) Physical description (size and weight)

(3) The effect (if any) on aerodynamic drag of the target

(4) Installation and performance data

(5) Frequency allocation data in accordance with 3.4.33

3.4.33 FREQUENCY ALLOCATION DATA. - The contractor shall submit complete frequency Allocation and Equipment Spectrum Signature Data in accordance with WR-29 for each new equipment or system which will require use of the radio frequency spectrum both active (transmitter) and passive (receiver). Preliminary data are required early in the design of the equipment with revised and complete data being submitted as it becomes available in accordance with WR-29. Submission of the frequency allocation data for NAVAIR use does not relieve the contractor of any responsibility of obtaining FCC authorization to operate the equipment while in his possession.

3.4.34 AIRCRAFT ARMAMENT CHARACTERISTICS REPORT. - This report shall be in accordance with Specification SAR-309.

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3.4.35 LAUNCHER INSTALLATIONS REPORT. - The report shall describe all target system components required for installation on the launcher and shall include weight, space, power, heating and cooling.

3.4.36 WEIGHT AND BALANCE DATA. - In accordance with the requirements of MIL-W-25140 or MIL-W-3947 as specified with submission of data as follows:

(1) Estimated Weight Report

(2) Weight and Balance Status Reports

(3) Calculated Weight Reports

(4) Actual Weight Reports and Appendices

NOTES: (a) For preparation of weight reports the contractor shall use the applicable pages of the weight statements, Parts I and II of MIL-STD-254 or MIL-STD-176 as applicable, supplemented by additional pages prepared by the contractor as may be necessary.

(b) In the requirements of 3.7.1, 3.7.6 and 3.7.7 of MIL-W-25140, substitute "free flight test portions of the contractor's formal demonstration of MIL-D-18243" in lieu of the words "Navy Parts II and III demonstrations and the trials".

3.4.37 PROPULSION SYSTEM DATA. - Propulsion system data shall include the installation of complete propulsion system, propellant systems, lubricating system, control system, cooling system, (including an airframe provided exhaust system,if required), induction system, starting system, and installation drawings of propellant tanks, coolant tanks, oxidizer tanks and pressure tanks. This includes detail design and installation drawings for solid propellant rockets where applicable. Propulsion system data shall also include: Detail and schematic drawings, sketches, substantiating calculations and test data of any propulsion system accessories, including pumping system, metering system, starting systems, secondary power system, etc., peculiar to the propulsion unit installation and which are developed by the contractor and not previously approved. Propulsion system data shall include the performance data of all accessories of the propulsion system necessary to define the gross propulsion capability and to substantiate original design calculations and objectives. For turbo-jet, turbo-prop, pulse-jet or ram-jet engine installations, a report of calculations of duct losses and installed performance corrections in accordance with MIL-D-17984 shall be submitted.

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3.4.38 TEST PROCEDURE MANUAL. - A test procedure manual shall be submitted which sets forth the plan and procedures for conducting the tests required by 3.5.1 and 3.5.3. The manual shall be in sufficient engineering detail to show specification compliance but need not extend to specific operations performed as a part of the manufacturing process. A suitable method shall be employed of cross referencing between the procedures described and the particular paragraphs of the applicable specifications with which they are designed to show compliance. The method of showing compliance with specification paragraphs for which there is no test procedure shall be stipulated. Sampling rates shall be specified where appropriate and shall include the sampling requirements of applicable specifications. Changes to the Test Procedure Manual stemming from design changes to applicable equipment must receive approval of NAVAIR prior to final adoption. Pending NAVAIR approval or disapproval, the contractor may make interim use of the changed procedures with the concurrence of the local Government representative.

3.4.39 DEVELOPMENT TESTS REPORTS. - A report shall be submitted for each group of tests required by 3.5.1. This report shall present the results of the tests, the contractor's analysis of results, including conclusions in sufficient engineering detail to enable NAVAIR to assess compliance with existing specifications. When detailed coverage is presented in other design data submitted under other provisions of this specification it need not be duplicated in this report. In such cases reference shall be made to the applicable report for additional information.

3.4.40 EXPERIMENTAL FLIGHT TEST REPORT. - Individual reports of the following types are required. NAVAIR may disapprove release of any specific target for flight when the flight does not appear to be in the best interest of the target system development.

3.4.40.1 PREFLIGHT PLANNING REPORT. - This report shall be prepared for each block of targets to be launched and shall include the data specified below. This report shall be amended by letter as found necessary prior to flight of a revised target system.

(1) Specific description of plans and objectives of flight.

(2) Specific description of equipment including block diagrams, wiring diagrams, general arrangement, test equipment, as changed from previous equipments.

(3) Deviations from the general flight test program plan as established in 3.4.2.

(4) Description of reliability tests to be performed prior to launching.

(5) Description of instrumentation required.

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3.4.40.2 PREFLIGHT DETAILED FLIGHT PLAN FOR EACH FLIGHT. - The plan shall include data sufficient to allow the test activity to plan for each flight, including:

(1) General description of plans and objectives of flight.

(2) Specific schedules.

(3) Specific instrumentation.

3.4.40.3 POST FLIGHT LAUNCHING LETTER REPORT. - Required within 48 hours after completion of flights, except when additional delay is essential to the presentation of data. The letter report shall be submitted after every target launching and shall be a brief description of the flight tests and results including:

(1) Statement of the Plans and Objectives.

(2) Visual observations.

(3) Preliminary conclusions.

3.4.40.4 POST FLIGHT EVALUATION REPORT. - This report shall be an appendix to the comparable preflight planning report (3.4.40.1) and shall contain:

(1) Analysis of flight results including comparison with expected results.

(2) Conclusions.

(3) Plans for feedback of conclusions into target system design.

3.4.41 HANDBOOK OF OPERATION AND SERVICE INSTRUCTIONS WITH PARTS BREAKDOWN. - This handbook shall be furnished by the contractor as a preliminary handbook, embodying a Major Component Parts List in lieu of an Illustrated Parts Breakdown, in accordance with the provisions of the contract design specification and MIL-H-7937. Handbooks shall be produced in white print form and furnished in accordance with schedule in Table I.

3.4.42 SAFETY DOCUMENTATION. - A safety manual or a safety chapter in the Mission Planning and Operational Manual (see 3.4.51) shall be provided in accordance with WS-4613.

3.4.43 SPECIFICATIONS AND SPECIFICATION REVISION PAGES. - Specifications and specification revision pages shall be submitted as follows:

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3.4.43.1 CONTRACTOR PREPARED SPECIFICATIONS. - The contractor shall submit specifications in the format designated by Standardization Manual M-200 and as specified below, for the following equipment:

(1) Target propulsion system

(2) Target component systems as appropriate in accordance with XAV format.

(3) Each item of launching aircraft equipment or logical grouping thereof in accordance with XAV format.

(4) Each item of operational test and checkout equipment or logical grouping thereof in accordance with XAV format.

(5) Each item of maintenance test equipment or logical grouping thereof in accordance with XAV format.

(6) Each item of handling equipment or logical grouping thereof provided by the contractor.

(7) Packaging (Shipping Containers)

3.4.43.2 SPECIFICATION REVISION PAGES. - Revision pages shall be furnished for the contract detail specification and for contractor prepared specifications of 3.4.43.1 in accordance with Naval Weapon Requirements WR-8 except for the following (the following paragraph numbers refer to those of WR-8).

3.2.1 (1) Add at the end of paragraph "or local Government representative".

3.2.1 (3) Change to read, "Guaranteed Performance and Performance Conditions. - Retain contract detail specification paragraphs unless modified under the provisions of 3.2.1(1) and add additional alternate paragraphs or performance figures as necessary to reflect the contractor's latest actual or estimated performance and weights. Additional paragraphs or figures shall be identified accordingly".

5. Delivery. Change to read, "Delivery. - Revised pages and indices shall be delivered at specified intervals and in the quantities specified in the addendum to MIL-D-8685 as applicable via the local Government representative. Classified revised pages and indices shall be handled in accordance with existing security regulations."

6.2.6 Aircraft. - Delete and substitute, "For purposes of this specification the term aircraft shall mean airplanes, helicopters, airships, guided missiles and targets."

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The final submittal shall represent the last article of the contract and shall include a cover page labeled "Final Corrected".

3.4.44 LOGBOOK. - A logbook with provisions for recording significant occurrences throughout the expected operating life of the target shall be provided for each target. A primary consideration in the design of the logbook shall be its suitability for use by Fleet operating personnel; however, the inclusion of supplementary provisions for recording detailed engineering data shall be permissible on logbooks delivered with developmental targets. Each logbook shall include:

(1) A transfer record containing information on shippers and receivers.

(2) A record sheet on target components. The types of data to be included are:

(a) Inventory of major serialized units installed in the target at time of acceptance.

(b) Field removal and replacement of major serialized units, and reason for replacement.

(3) A record sheet on performance of standard field preparation checks, with operating time and target status at end of the check.

(4) A record sheet of launching attempts, flight results and target recovery (briefly abstracted).

(5) A record sheet of engineering changes/incorporations or technical order compliances.

(6) A statement providing field activities with information on implementation of the warranty or guarantee provisions of the production contract.

(7) Disposition instructions of expended target logbooks should be provided.

3.4.45 DEMONSTRATION PLAN. - A Formal Demonstration plan shall be submitted. The plan shall include:

(1) Indication by the contractor that the target system is ready for demonstration and that it will meet the performance requirements of the applicable detail specification.

(2) Test schedules for demonstrating items required by the applicable addendum to MIL-D-18243.

(3) Methods for obtaining, processing, and analyzing data.

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(4) Methods of presenting results.

(5) A list of materials and services required specifically for this demonstration.

3.4.46 DEMONSTRATION REPORT. - A demonstration report shall be submitted. The report shall summarize the results of the formal demonstration including:

(1) Discussion of all items required to be demonstrated by the applicable demonstration addendum.

(2) Discussion of the ability of the target to fly predetermined flight paths.

(3) A comparison of the required versus achieved performance.

(4) A comparison of the required versus achieved reliability.

(5) The reliability of the target system derived from the data collected during the demonstration program.

(6) The time, manpower and facilities required to prepare the target for flight determined from data collected during the demonstration.

(7) A complete analysis of all failures and defects which occurred in the demonstration, their causes, and their corrections.

3.4.47 ALTERATION REPORT. - DEMONSTRATION TARGETS. - An individual report on each alteration (alteration shall be as defined in MIL-D-18243) proposed for incorporation in the demonstration targets. Each report shall include the following information:

(1) Number of alteration (numerical sequence in any particular demonstration)

(2) Detailed description of alteration

(3) Reason for alteration

(4) Serial numbers of targets affected

(5) Affect on validity of tests already completed

(6) Contractor's signature

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3.4.48 MICROFILM AND TABULATING CARDS. - Microfilm and tabulating cards of all engineering drawings and associated lists shall be furnished in accordance with MIL-C-9878/1 and the applicable completed NAVWEPS Form 4200/25.

3.4.48.1 PREPARATION OF ENGINEERING DRAWINGS. - Engineering drawings and associated lists for the microfilm of 3.4.48 shall be prepared in accordance with MIL-D-1000, MIL-D-1000/1, and a completed NAVWEPS Form 4200/25.

3.4.49 INTERCHANGEABILITY DATA. - The contractor shall provide the replaceability interchangeability data required for the target, using applicable portions of MIL-I-8500.

3.4.50 RADAR REFLECTIVITY REPORT. - A report shall be submitted describing the actual radar reflectivity characteristics of the target. Included in the report shall be:

(1) A description of the radar augmentation system used on the target.

(2) Radar cross section patterns as a function of frequency, polarization, bistatic angle, aspect angle with and without augmentation as applicable.

(3) Specific measurement parameters for patterns shall be specified by NAVAIR.

(4) For active augmentation systems the AM and FM noise characteristics of the active augmentor shall be submitted as well as such characteristics as noise figure, power output, gain, etc.

3.4.51 MISSION PLANNING AND OPERATIONAL MANUAL. - This manual shall be furnished by the Contractor and shall include sufficient data to allow the user activities, the tactical surface and air arms, to successfully employ their exercise weapons. This data shall delineate target launch characteristic, performance, and the Target Systems radar augmentation and miss-distance measuring systems. Particular emphasis shall be placed on accurately describing the expected radar reflectivity for the various surface-to-air search, acquisition, and both bistatic and monostatic guidance radars of the Guided Missile Ships of the Fleet and the applicable radars of the Army, Navy, Air Force and Marine Corps air arm. MIL-M-7700, MIL-H-5474, and in general accordance with MIL-M-81273 shall be used as a general guide in preparing this manual.

3.4.52 QUALITY ASSURANCE DOCUMENTATION. - Quality Assurance Provisions shall be prepared in accordance with WR-43.

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3.4.53 SUMMARY OF DESIGN DATA. - The contractor shall submit a complete list in report form of the estimated dates for submittal of design data required by paragraph 3.4 except for structural items which are covered in paragraph 3.4.16. This report shall consist of the titles and submittal dates of all design data. This list shall reference contractor's letters by which data have been submitted and the letters by which data have been accepted including the serial number (if any), the date and the correspondence designation (i.e. AIR-520, AIR-535, AIR-510 etc.) and shall show estimated submittal dates for data not yet submitted. The summary shall list each paragraph in the applicable contract addendum and shall reference the letters which forwarded data under that paragraph.

3.5 TESTS REQUIRED. - Contractors not having laboratory testing facilities satisfactory to the Government Inspector shall engage the services of a commercial testing laboratory acceptable to NAVAIR. The following tests are required:

3.5.1 DEVELOPMENT TESTS. - Development tests shall be performed by the contractor and shall be conducted in accordance with the Test Procedure Manual of 3.4.38. Development Tests shall include all tests required to determine that the target system and its major components meet the requirements of the applicable specifications. The right is reserved by NAVAIR or the LGR to modify the tests or require any additional tests deemed necessary to determine compliance with the applicable specifications. Development tests shall include, but not necessary be limited to the following:

(1) All ground, laboratory and flight tests incident to the selection of components and investigation of the performance of breadboard or engineering models of component systems, subassemblies or assemblies under ambient and environmental conditions.

(2) Electronic Equipment Tests - In accordance with MIL-T-5422.

(3) Radio and Electronic Interference Tests - In accordance with MIL-E-6051, MIL-I-6181 and MIL-I-16910 as applicable.

(4) Armament System Tests - In accordance with test procedures approved by NAVAIR.

(5) Structural Tests - Structural tests are required for target, pylon and launcher to failing loads. In addition, proof tests to 1.1 times limit load are required at installation in the test airplane used for development of the target unless tests already performed by the airplane manufacturer have proven the strength of the airplane. If the effects of aerodynamic heating are significant, structural and flutter tests shall be performed in the heat facility at the Aeronautical Structures Laboratory, NAEC, Philadelphia. Tests shall be in accordance with test procedures approved by NAVAIR.

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(6) Propulsion System Tests. - The following tests shall be conducted by the contractor:

(A) Propellant System Tests. - Propellant System tests in accordance with MIL-F-17874 and MIL-I-18802 as applicable, except that flight tests are not required.

(B) Propulsion System Accessories Test. - Propulsion system accessories tests on all power plant accessories developed and furnished by the contractor, i.e., power plant accessories not contractor-furnished equipment (approved type).

(7) Hydraulic or Pneumatic System Tests - In accordance with test procedures approved by NAVAIR.

(8) Electrical System Tests - In accordance with test procedures approved by NAVAIR.

(9) Ice Protection and Temperature Control System Tests - In accordance with test procedures approved by NAVAIR.

(10) Shipboard vibration tests - Tests shall be conducted in accordance with MIL-STD-167(SHIPS). In addition these tests shall determine the maximum shock that the target can withstand when stowed in a manner similar to that in which it will be stowed when aboard ship.

(11) Safety Tests - In accordance with test procedures approved by NAVAIR.

(12) Aerodynamic Tests - Performance and stability and control tests in accordance with test procedures approved by NAVAIR.

(13) Miscellaneous - Other tests required by the LGR or which in the opinion of the contractor are required to fully test the article.

3.5.2 RELIABILITY TESTS. - Reliability tests shall be conducted in accordance with the Reliability Report of 3.4.3 as approved by NAVAIR.

3.5.3 ACCEPTANCE TESTS. - Acceptance Tests shall be conducted in accordance with the Test Procedure Manual of 3.4.38 as approved by NAVAIR, by the contractor and witnessed by the LGR for all items submitted for acceptance under the contract. Acceptance tests shall consist of Individual Tests, Sampling Tests and Special Tests:

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(1) Individual Tests - Individual Acceptance tests shall be conducted on each item submitted for acceptance under the contract.

(2) Sampling Tests - Sampling acceptance tests shall be conducted on a quantity of items as specified in the applicable specifications and/or the Test Procedure Manual of 3.4.38 as approved by NAVAIR. Items for sampling acceptance tests shall be selected by the LGR from those items which have passed the Individual Tests.

(3) Special Tests - Special acceptance tests shall be conducted on the first item delivered after any engineering change. These tests shall be commensurate with the scope of the change and shall be used in cases where the scope does not warrant repeating the tests of 3.5.1, above. The tests shall be conducted in accordance with the Test Procedure Manual of 3.4.38 as approved by NAVAIR.

3.5.4 SERVICE TESTS. - Service tests will be performed by the Government on equipment provided for that purpose under the contract to verify that the items delivered satisfy the requirements of the applicable specifications. The results of the service tests may be used by the Government as evidence affecting the acceptance of like articles under the contract.

3.5.5 WITNESSING OF TESTS. - Acceptance tests performed by the contractor shall be witnessed by the LGR unless otherwise directed by NAVAIR.

LGR may request from NAVAIR the assistance of other NAVAIR field activities as required. Flight tests performed at the contractor's plant shall be witnessed by the LGR. Flight tests performed at a NAVAIR test activity shall be witnessed by the agency designated by NAVAIR.

4. QUALITY ASSURANCE PROVISIONS.

4.1 SAMPLING. - There are no applicable requirements.

4.2 EXAMINATION AND ACTION.

4.2.1 EXAMINATION. - The LGR shall examine all design data for completeness, and for compliance with applicable specifications and contractural requirements. Design data not conforming to applicable requirements shall be returned to the contractor for revision prior to the distribution of Table I. For data submitted to NAVAIR the LGR shall forward his comments by endorsement on the contractor's forwarding letter.

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4.2.1.1 SUMMARY OF DESIGN DATA. - The LGR shall carefully inspect the Summary of Design Data for agreement with the records of his office and comment accordingly when forwarding the Summary and revisions thereof.

4.2.2 ACTION. - Action taken upon design data and tests shall be by acceptance, or review (information only) as specified in Table I. Action taken on drawings shall be by release or review (information only) as specified in Table I.

4.2.2.1 COMPLETENESS OF DESIGN DATA AND DRAWINGS. - In the acceptance and release of design data and drawings NAVAIR or the LGR does not check or assume responsibility for accuracy or completeness of details or for any deviation from applicable specifications unless specifically approved by NAVAIR. All authorized deviations from the contract detail specification and other applicable specifications shall be specifically brought to the attention of NAVAIR in the letter forwarding the data.

4.2.2.2 REVISION OF DESIGN DATA. - Design data and drawings, requiring revision and resubmittal for acceptance or release by NAVAIR or the local Government representative shall have all corrections so marked that they may be readily found. Design data, once accepted or released by NAVAIR or the LGR shall not be changed without written authorization by NAVAIR or the LGR as applicable, and when so authorized, replacement copies shall be furnished to all addressees.

4.2.2.3 RELEASE OF DRAWINGS. - After acceptance or release of data submitted as required by this specification or addenda thereto, the LGR shall release all further drawings necessary for manufacture which do not conflict with or change what has already been accepted or released. The intent of this provision is that once the major features of design are released, the manufacturing drawings will be developed in accordance therewith. Release of drawings shall be signified by the dated signature, or initialing by the LGR in the drawing block reserved for Government approval. Such release of any drawing will constitute official release for production or procurement for service use of the item, or items described thereon. Release of drawings does not constitute official release of unauthorized deviations from detail or design specification requirements and shall not be construed as acceptance of non-government specifications or standards. In releasing manufacturing drawings for fabrication of parts, the LGR, as a representative of NAVAIR, shall not assume responsibility for accuracy, completeness of detail, or proper fit or functioning of contractor-constructed parts or assemblies. The responsibility for satisfactory construction and functioning of the target system or target system parts, and for satisfactory stability and weight, rests entirely with the contractor.

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4.2.2.4 DESIGN DATA SUBMITTED FOR INFORMATION. - Design data submitted to NAVAIR for information may not be acknowledged, however, NAVAIR may notify the contractor of any objectionable design features, etc., which would not be satisfactory for future target systems.

5. PREPARATION FOR DELIVERY.

5.1 DESIGN DATA SUBMITTAL. - Design data for target systems shall be submitted as specified in Table I. The contractor shall make every effort to expedite submittal of design data in order to allow sufficient time for review by NAVAIR prior to specific check points or release for flight, as applicable.

5.2 METHOD OF SUBMITTAL. - Contract design data, submitted to NAVAIR shall be via the LGR in the quantities specified herein. All letters forwarding design data and other correspondence relative to design data that have been submitted, shall be submitted via the LGR in triplicate. Design data of unrelated subject matter, for example: Structural Design Data and Electrical Installation Drawings, or Instrument Drawings and Power Plant Drawings, shall not be submitted under the same forwarding letter. Design data shall be shipped at the contractor's expense.

6. NOTES.

6.1 RESPONSIBILITY FOR APPLICABLE SPECIFICATION AND PUBLICATIONS. - LGR's are furnished copies of the latest issue of all applicable specifications and publications as they become available. The responsibility for ascertaining and following the revisions of specifications applicable to a specific contract rests with the contractor. NAVAIR or LGR's office will inform contractors of the number and date of the latest issue of any specification upon request.

6.2 DUPLICATION OF DATA. - Duplication of data shall be avoided. Data should be referenced when included in previous reports submitted under this addendum and required in subsequent reports.

6.3 REVISION OF DATA. - Revision to reports submitted in accordance with Table I shall be made whenever new information invalidates such reports.

6.4 SECURITY CLASSIFICATION. - Classified design data and drawings shall contain the proper security classification on drawings and on each page of reports, specifications, photographs, etc. in accordance with existing security regulations.

6.5 USE OF THIS SPECIFICATION. - This general specification shall be used as the standard form for the preparation of addenda. Addenda shall be prepared as follows:

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6.5.1 Agree with this specification in paragraph arrangement, numbering and headings, except that where a paragraph is listed in the addendum as "applicable", "not applicable" or "not required" subsequent subparagraphs may be omitted provided numbering sequence is not affected.

6.5.2 Unless paragraphs of addenda completely supersede corresponding paragraphs of this specification, paragraphs will be labelled as "applicable" or "required" (with or without specific deviations or supplementary requirements), "not applicable", or "not required". General requirements or instructions will be designated as "applicable" or "not applicable", and requirements for specific data or action will be designated as "required" or "not required".

6.5.3 Paragraphs and subparagraphs will be added as required.

6.6 DEFINITIONS.

6.6.1 DIFFERENCES OVER PROTOTYPE. - The term "differences over prototype" when used in addenda to this specification shall be defined to mean that the design data required for a target system for which there is a specified prototype shall consist of only such additional data as are necessary to reflect the differences over the data submitted for the previous design. In each case the applicable prototype will be specified in the addendum.

6.6.2 ABBREVIATIONS. - Identifying abbreviations used herein are defined as follows:

NAVAIR -NAVAL AIR SYSTEMS COMMAND, HEADQUARTERS Washington, D. C. 20360

LGR -Local Government Representative - Designated Government representative having responsibility for contract administration services at the contractor's plant.

NAVMISCEN -Naval Missile Center, Point Mugu, California 93041

NADC -Naval Air Development Center, Johnsville, Warminster, Pennsylvania 18974

NAMC -Naval Air Material Center, Phila., Pa. 19112

NAEL(SI) -Naval Air Engineering Laboratory (Ship Installation), Naval Air Engineering Center, Phila., Pa. 19112

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NPC	-Naval Photographic Center, Naval Air Station, Washington, D. C. 20390
ASO	-Aviation Sypply Office, Phila., Pa. 19111
DTMB	-David Taylor Model Basin, Washington, D. C. 20007
ONR	-Office of Naval Research, Washington, D. C. 20390
NAVAIRSYSCOMREPCEN	-Naval Air Systems Command Representative, Central, Wright-Patterson Air Force Base, Ohio 45433
NAVAIRSYSCOMREPLANT	-Naval Air Systems Command Representative, Atlantic, Naval Air Station, Norfolk, Virginia 23511
NAVAIRSYSCOMREPPAC	-Naval Air Systems Command Representative, Pacific, Naval Air Station, North Island, San Diego, Calif. 92135
COMNAVAIRPAC	-Commander, Naval Air Force, U.S. Pacific Fleet, Naval Air Station, San Diego, California 92134
COMNAVAIRLANT	-Commander, Naval Air Force, U.S. Atlantic Fleet, Naval Air Station, Norfolk, Virginia 23511
NAFI	-Naval Avionics Facility, Indianapolis, Indiana 46218
NSD	-Naval Supply Depot, 5801 Tabor Ave., Phila., Pa. 19120
NATC	-Naval Air Test Center, Patuxent River, Maryland 20670
NWEF	-Naval Weapons Evaluation Facility, Albuquerque, New Mexico 87117
DDC	-Defense Documentation Center for Administrator of Scientific Technical Information, Bldg. No. 5, Cameron Station, Alexandria, Virginia 22314
NWL	-Naval Weapons Laboratory, Dahlgren, Virginia 22448

6.7 DEVIATIONS. - Deviations from this specification shall not be permitted unless specified in addenda to this specification, in the contract amendment, or by other written authorization of NAVAIR.

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TABLE I

UNLESS OTHERWISE SPECIFIED IN THE ACTION COLUMN ALL DATA AND TESTS SHALL BE ACCEPTED OR RELEASED BY THE LOCAL GOVERNMENT REPRESENTATIVE

Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.1	Program Master Plan	AIR-535 NAVMISCEN NADC	Accept. Info. Info.	4 Nonrepro. 4 " 3 "	Recurring	Not later than 90 days after notification of contract award. Revision pages quarterly thereafter.
3.4.2	Development Program Planning Report	AIR-535 NAVMISCEN NADC NAEL SI ASO	Accept. Info. Info. Info. Info.	4 Nonrepro. 4 " 1 " 1 " 1 "	Recurring	Not later than 90 days after notification of contract award, then as program changes.
3.4.3	Reliability Reports	AIR-535 NAVMISCEN NADC ASO	Accept. Info. Info. Info.	14 Nonrepro. 2 " 3 " 1 "	Recurring	As per MIL-STD-1304
3.4.4	Environmental Criteria Report	AIR-535 NAVMISCEN NADC NAEL(SI)	Accept. Info. Info. Info.	3 Nonrepro 2 " 3 " 1 "	Recurring	Not later than 60 days after each series of tests performed
3.4.5	Target System Brochure	AIR-535 NADC NAEL SI ASO AIR-604 DDC	Accept. Info. Info. Info. Info. Info.	20 Photo & 3 Offset 1 " 1 " 2 " 20 "	Recurring	Not later than 6 months after notification of contract award.
3.4.6	Program Evaluation and Review Technique Reporting	AIR-535	Info.	2 Nonrepro.	Recurring	Not later than 45 days prior to initial Target System Review, then as program progresses.

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TABLE I (Cont)

Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.7	Monthly Progress Letters	AIR-535 NADC NAEL SI LGR	Info. Info. Info. Info.	10 Nonrepro. 3 " 1 " 1 "	Recurring	Not later than 15 days after the end of each month.
3.4.8(2)	Mockup Photographs	AIR-535 NADC NAEL SI ASO LGR	Info. Info. Info. Info. Info.	3 Sets 1 Prints 1 " 1 " 1 "	Non-Recurring	Not later than 15 days after Mockup inspection.
3.4.9.1	Slides, Charts, Graphs, etc.	AIR-535	Info.	2 Nonrepro.	Recurring	Not later than 30 days subsequent to request by NAVAIR
3.4.9.2	Motion Picture Film	NPC AIR-535	Accept. Info.	1 Orig. 1 Copy	Recurring	As specified in 3.4.9.2
3.4.10	Simulator Program Reports	AIR-535 AIR-604 NADC DDC	Accept. Info. Info. Info.	4 Nonrepro. 2 " 1 " 20 "	Non-Recurring	Not later than 60 days after completion of a major program
3.4.11	Aircraft Installation Report	AIR-535 NAVMISCEN NADC NAEL SI	Accept. Info. Info. Info.	4 Nonrepro. 3 " 3 " 1 "	Recurring	Not later than 60 days after Mockup. Revisions as program progresses
3.4.12	Ship Installation Report	AIR-535 NAVMISCEN NADC NAEL SI	Accept. Info. Info. Info.	4 Nonrepro. 3 " 3 " 1 "	Recurring Final	Not later than nine months after Mockup. Revisions as the program progresses. Not later than 60 days after completion of demonstration.

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TABLE I (Cont)

Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.13	Handling Equipment Report	AIR-535 NAVMISCEN NADC NAEL SI NWEF	Accept. Info. Info. Info. Info.	4 Nonrepro. 3 " 3 " 1 " 1 "	Recurring	Not later than 60 days after Mockup. Revisions as the program progresses.
3.4.14	Shipping Containers Report	AIR-535 NAVMISCEN NADC NAEL SI	Accept. Info. Info. Info.	2 Nonrepro. 3 " 3 " 1 "	Non-Recurring	Not later than 30 days prior to Initial Target System Review.
3.4.15.1.1	Aerodynamic Investigation Program Report	AIR-535 NAVMISCEN NADC	Accept. Info. Info.	3 Nonrepro. 3 " 3 "	Recurring	Not later than 30 days after notification of contract award. Revisions prior to undertaking any changes in the test program
3.4.15.1.1(1)	Proposed Investigation Programs Report for Tests at Government-Furnished Facilities	Testing Facility AIR-535 NADC	Info. Accept. Info.	3 Nonrepro. 3 " 3 "	Non-Recurring	As soon as practicable, but not later than 30 days prior to requested test starting date.
3.4.15.1.1(2)	Investigation Programs Report for tests at contractor's or other privately operated facilities.	AIR-535 NADC	Info. Info.	3 Nonrepro. 3 "	Non-Recur-	Not later than 30 days prior to scheduled test starting date.

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TABLE I (Cont)

Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.15.1.2	Aerodynamic Model Drawings	Testing Facility	Accept.	2 Nonrepro.	Non-Recurring	Prior to starting model construction and as soon as practicable following test facility acceptance of design.
		Construction Facility	Info.	2 "		
3.4.15.1.3(1)	Interim Letter Reports	AIR-535	Info.	3 Nonrepro.	Non-recurring	Not later than 20 days after completion of testing during each facility occupancy.
		NAVMISCEN	Info.	3 "		
		NADC	Info.	3 "		
3.4.15.1.3(2)	Test Data Reports	AIR-535	Info.	3 Nonrepro.	Non-recurring	Not later than 40 days following receipt of test data by contractor.
		NAVMISCEN	Info.	3 "		
		NADC	Info.	3 "		
3.4.15.2.1	Aerodynamic and Thermodynamic Heating Analysis Report	AIR-535	Accept.	4 Nonrepro.	Non-recurring.	Not later than 6 months prior to estimated date of first inflight launching.
		NAVMISCEN	Info.	2 "		
		NADC	Info.	3 "		
		AIR-604	Info.	2 "		
		DDC	Info.	20 "		
3.4.15.2.2	Structural Heating Summary Report	AIR-535	Accept.	4 Nonrepro.	Non-recurring	Not later than 30 days prior to first flight
		NAVMISCEN	Info.	2 "		
		NADC	Info.	3 "		
		AIR-604	Info.	2 "		
		DDC	Info.	20 "		
3.4.15.2.3	Component Heating Summary Report	AIR-535	Accept.	5 Nonrepro.	Non-recurring	Not later than 30 days prior to first flight
		NAVMISCEN	Info.	2 "		
		NADC	Info.	3 "		
		AIR-604	Info.	2 "		
		DDC	Info.	20 "		

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TABLE I (Cont)

Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.15.3	Stability and Control Report	AIR-535	Accept.	4 Nonrepro.	Recurring	Initial Report to be submitted not later than 90 days after notification of contract award. Subsequent revisions or replacements at intervals not greater than four months or a statement to the effect that no revisions have been made shall be submitted.
		NAVMISCEN	Info.	3 "		
		NADC	Info.	3 "		
		*AIR-604	Info.	2 "		
		*DDC	Info.	20 "		
	*Final report only.				Final	Not later than 30 days prior to first free flight.
3.4.15.4	System Dynamics Report	AIR-535	Accept.	4 Nonrepro.	Recurring	Initial report to be submitted not later than 90 days after notification of contract award. Subsequent revisions or replacements at intervals not greater than four months or a statement to the effect that no revisions have been made shall be submitted.
		NAVMISCEN	Info.	3 "		
		NADC	Info.	3 "		
					Final	Not later than 30 days prior to first free flight.

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TABLE I (Cont)

Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.15.5.1	Basic Data	AIR-535 NADC	Accept. Info.	3 Nonrepro. 3 "	Recurring	Not later than 90 days after notification of contract award. Revisions whenever configuration changes result in significant performance changes.
3.4.15.5.2(1)	Performance Data Report	AIR-535 NAVMISCEN NADC NAEL SI	Accept. Info. Info. Info.	3 Nonrepro. 3 " 3 " 1 "	Recurring	Not later than 30 days after approval of basic data. Revisions whenever configuration changes result in significant performance changes.
3.4.15.5.2(2)	SAC Charts	AIR-535	Accept.	2 Repro. glossy prints	Recurring	Same as (1) above.
3.4.15.5.3	Additional Submittals	Same as for 3.4.15.5.2(1)				Not later than 60 days following receipt of request from NAVAIR.
3.4.15.6	Launching Characteristics	AIR-535 NAVMISCEN NADC NAEL SI	Accept. Info. Info. Info.	6 Nonrepro. 3 " 3 " 1 "	Recurring	Not later than 30 days prior to Initial Target System Review. Revisions as program progresses.

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Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.16	Structural Design Analyses and Test Data	AIR-535 NAVMISCEN NADC	Accept. Info. Info.	1 Nonrepro. 4 " 3 "		Submittal Date as specified in MIL-A-8868 and MIL-A-8870, as modified by 3.4.16.
3.4.16.1	Structural Drawings	AIR-535 NAVMISCEN NADC	Accept. Info. Info.	1 Nonrepro. 1 " 1 "		Not later than 30 days prior to release for manufacturing.
3.4.17.1	Information on Non-Standard Parts	AIR-535 NAVMISCEN NADC NAFI	Info. Info. Info. Accept.	2 Nonrepro. 1 " 1 " 2 "	Non-recurring	Not later than 60 days after mockup.
3.4.17.2	Nomenclature Nameplates and Serial Numbers	AIR-52021	Accept.	2 Nonrepro.	Non-recurring	Not later than 60 days after mockup.
3.4.17.3	Schematic Diagrams	AIR-535 NAVMISCEN NADC	Release Info. Info.	2 Nonrepro. 3 " 3 "	Non-recurring	Not later than 30 days prior to Initial Target System Review.
3.4.17.4	Master Wiring Diagrams	AIR-535 NAVMISCEN NADC	Release Info. Info.	2 Nonrepro. 3 " 3 "	Non-recurring	Not later than 30 days prior to Initial Target System Review.
3.4.17.6	Electrical Load Analysis	AIR-535 NAVMISCEN NADC	Release Info. Info.	2 Nonrepro. 2 " 3 "	Non-recurring	Not later than 30 days prior to Initial Target System Review.
3.4.18	Materials and Processes Development & Evaluation Report	AIR-535 NAVMISCEN NADC AIR-604 DDC	Info. Info. Info. Info. Info.	2 Nonrepro. 1 " 3 " 2 " 20 "	Recurring	Not later than 30 days prior to Initial Target System Review, then Semi-annually.

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Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.19	Magnesium Alloy Parts List	AIR-535	Info.	2 Nonrepro.	Recurring	Prior to fabrication of any parts of the materials concerned then as changes occur.
		NAVMISCEN	Info.	1 "		
		NADC	Info.	3 "		
3.4.20	Finish Specification	AIR-535	*Info.	1 Nonrepro.	Nonrecurring	Not later than 60 days after mockup.
		LGR	Accept.	5 "		
		NAVMISCEN	Info.	1 "		
		NADC	Info.	3 "		
	*Acceptance if deviations are taken.					
3.4.21	Heating and Cooling Report	AIR-535	Accept.	2 Nonrepro.	Nonrecurring	Not later than 30 days prior to Initial Target System Review.
		AIR-604	Info.	2 "		
		NAVMISCEN	Info.	2 "		
		NADC	Info.	3 "		
		NAEL SI	Info.	1 "		
		DDC	Info.	20 "		
3.4.22	Hydraulic System Data	AIR-535	Accept.	3 Nonrepro.	Recurring	Not later than 30 days prior to Initial Target System Review. Revisions as Program progresses.
		NAVMISCEN	Info.	1 "		
		NADC	Info.	3 "		
		NAEL SI	Info.	1 "		
3.4.23	Pneumatic System Data	AIR-535	Accept.	3 Nonrepro.	Recurring	Not later than 30 days prior to Initial Target System Review. Revisions as Program progresses.
		NAVMISCEN	Info.	3 "		
		NADC	Info.	3 "		
		NAEL SI	Info.	1 "		

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TABLE I (Cont)

Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.24	Recovery System Report	AIR-535	Accept.	3 Nonrepro.	Recurring	Not later than 30 days prior to Initial Target System Review. Revisions as program progresses.
		NAVMISCEN	Info.	1 "		
		NADC	Info.	3 "		
		NAEL SI	Info.	1 "		
3.4.25	Human Factors Preliminary Report	AIR-535	Info.	3 Nonrepro.	Non-recurring	Not later than four months after notification of contract award.
		NAVMISCEN	Info.	1 "		
		NADC	Info.	3 "		
	Human Factors Final Report	AIR-535	Accept.	3 Nonrepro.	Recurring	Not later than 60 days after Mockup. Revisions as the program progresses.
		AIR-604	Info.	2 "		
		NAVMISCEN	Info.	1 "		
		NADC	Info.	3 "		
		DDC	Info.	20 "		
3.4.26(1)	Information on Non-Standard Parts	AIR-535	Info.	2 Nonrepro.	Recurring	Not later than 60 days after Mockup. Revisions as program progresses.
		NAVMISCEN	Info.	1 "		
		NADC	Info.	1 "		
		NAFI	Accept.	2 "		
3.4.26(2)	Tube Complement Report	AIR-535	Info.	2 Nonrepro.	Recurring	Not later than 30 days prior to Initial Target System Review. Revisions as program progresses.
		NADC	Info.	1 "		
		NAFI	Info.	2 "		
3.4.26(3)	Nomenclature, Nameplates and Serial Numbers	AIR-52021	Accept.	2 Nonrepro.	Non-recurring	Not later than 60 days after Mockup.
3.4.26(4)	Design Approval Model Data	AIR-535	Info.	2 Nonrepro.	Non-recurring	To accompany model specified by NAVAIR.
		NAVMISCEN	Info.	1 "		
		NADC	Info.	1 "		

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Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.27	Control and Stabilization System Report	AIR-535 NAVMISCEN NADC	Accept. Info. Info.	4 Nonrepro. 3 " 3 "	Recurring	Not later than 30 days prior to Initial Target System Review. Revisions as program progresses.
3.4.28	Guidance System Report	AIR-535 NAVMISCEN NADC	Accept. Info. Info.	4 Nonrepro. 3 " 3 "	Recurring	Not later than 30 days prior to Initial Target System Review. Revisions as the program progresses.
3.4.29	Maintenance and Operational Test and Checkout Equipment Report	AIR-535 NAVMISCEN NADC NAEL SI ASO	Accept. Info. Info. Info. Info.	4 Nonrepro. 3 " 3 " 1 " 1 "	Recurring	Not later than 30 days prior to Initial Target System Review. Revisions as the program progresses.
3.4.30	Flight Test Equipment Report	AIR-535 NAVMISCEN NADC	Accept. Info. Info.	4 Nonrepro. 2 " 3 "	Recurring	Not later than 60 days after Mockup. Revisions as the program progresses.
3.4.31	Telemetering Electronic Design Data					
	Test Procedures	AIR-535 LGR NADC	Accept. of info. as determined by LGR	3 Nonrepro. 1 " 1 "	Recurring	Not later than 15 days prior to start of tests changes prior to use of contractor

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Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.31 (Cont)						
	Report on Contractor's Tests	AIR-535 LGR NADC	Info. Info. Info.	3 Nonrepro. 1 " 1 "	Non-recurring	Not later than 30 days after completion of tests
	Nomenclature	AIR-52021 AIR-535	Assign Info.	1 Nonrepro. 1 "	Recurring	As soon as possible and not later than 60 days prior to placing order for nameplates
	Specification	AIR-535 LGR NAVMISCEN	Info. Info. Info.	6 Nonrepro. 2 " 1 "	Recurring	Within 30 days of NAVAIR approval
	Frequency Allocation	AIR-535 NAVMISCEN	Accept. Info.	3 Nonrepro. 1 "	Non-recurring	Not later than 6 months prior to delivery of first equipment
3.4.32	Antenna Data Report	AIR-535 NAVMISCEN NADC	Accept. Info. Info.	2 Nonrepro. 2 " 3 "	Recurring	Not later than 60 days after Mockup. Revisions as the program progresses.
3.4.33	Frequency Allocation Data	AIR-535 NAVMISCEN	Accept. Info.	6 Nonrepro. 1 "	Recurring	Prior to fabrication of equipment. Revisions as changes occur
3.4.34	Aircraft Armament Characteristics Report	AIR-535 NADC	Accept. Info.	2 Nonrepro. 1 "	Recurring	Not later than 30 days after completion of design, then as changes occur in the design.

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TABLE I (Cont)

Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.35	Launcher Installations Report	AIR-535 NAEL SI NADC	Info. Info. Info.	4 Nonrepro. 1 " 1 "	Recurring	At least 60 days prior to first flight of equipment and revisions whenever changes in the installations are designed.
					Final	Not later than 60 days after completion of design.
3.4.36(1)	Estimated Weight Report	AIR-506 LGR NADC	Accept. Info. Info.	1 Nonrepro. 1 " 1 "	Non-recur-	Not later than 30 days after completion of the design phase represented by the report.
3.4.36(2)	Weight and Balance Status Reports	AIR-506 LGR NADC	Info. Info. Info.	1 Nonrepro. 1 " 1 "	Recurring	Not later than 15 days after completion of the design phase represented by the report.
3.4.36(3)	Calculated Weight Report	AIR-506 LGR NADC	Accept. Info. Info.	1 Nonrepro. 1 " 1 "	Non-recurring	Not later than 30 days after completion of the design phase represented by the report.
3.4.36(4)	Actual Weight Reports and Appendices	AIR-506 LGR NADC	Accept. Info. Info.	1 Nonrepro. 1 " 1 "	Non-recurring	Not later than 45 days after the article is weighed except targets for test not later than time of delivery to test activity.

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TABLE I (Cont)

Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.37	Propulsion System	AIR-535 NAVMISCEN NADC NAEL SI	Accept. Info. Info. Info.	4 Nonrepro. 3 " 3 " 1 "	Recurring	Not later than 30 days prior to Initial Target System Review. Revisions as program progresses
3.4.38	Test Procedure Manual	AIR-535 NADC NAEL SI	Accept. Info. Info.	4 Nonrepro. 3 " 1 "	Recurring	Not later than 90 days prior to intended use. Revisions not later than 30 days prior to their expected use.
3.4.40	Development Tests Reports	AIR-535 NADC	Accept. Info.	4 Nonrepro. 3 "	Non-recurring	Not later than 30 days after completion of tests and reduction of data.
3.4.40.1	Preflight Planning Report	AIR-535 NAVMISCEN NADC	Accept. Info. Info.	4 Nonrepro. 3 " 3 "	Recurring	At least 60 days prior to first target flight in each group of flight tests.
3.4.40.2	Preflight Detailed Flight Plan for each Flight	NAVMISCEN AIR-535	Accept. Info.	2 Nonrepro. 1 "	Recurring	As specified by NAVMISCEN.
3.4.40.3	Post-Flight Firing Letter Report	AIR-535 NAVMISCEN NADC	Info. Info. Info.	4 Nonrepro. 3 " 3 "	Recurring	Not later than 10 days after target flight.

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TABLE I (Cont)

Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.40.4	Post-Flight Evaluation Report	AIR-535	Info.	4 Nonrepro.	Recurring	Not later than 60 days after completion of test and reduction of data.
		NAVMISCEN	Info.	3 "		
		NADC	Info.	3 "		
3.4.41	Handbook of Operation and Service Instructions with Parts Breakdown	AIR-535	Info.	5 Nonrepro.	Final	Copies of the handbook shall be furnished in white print form and shall be delivered(except copies delivered with each article)at least 30 days before delivery of the first article.
		LGR	Info.	1 "		
		NAVMISCEN	Info.	3 "		
		NADC	Info.	1 "		
		NAMC	Info.	1 "		
		NAEL SI	Info.	1 "		
		Each article delivered 1 Nonrepro.				
3.4.42	Safety Documentation	AIR-535	Accept.	1 Nonrepro.	Final	At least 15 days prior to start of demonstration
		NADC	Info.	1 "		
		NAEL SI	Info.	1 "		
3.4.43.1	Contractor-Prepared Specifications	AIR-535	Accept.	4 Nonrepro.	Final	Not later than 30 days prior to Initial Target System Review.
		NADC	Info.	3 "		
		NAEL SI	Info.	1 "		
		ASO	Info.	2 "		
3.4.43.2	Revision pages to Contractor-prepared Specifications	AIR-535	Info.	4 Nonrepro.	Recurring	Beginning 120 days subsequent to Initial Target System Review and at 120-day intervals thereafter.
		NADC	Info.	3 "		
		NAEL SI	Info.	1 "		
		ASO	Info.	2 "		

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Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.43.2	Revision pages to contract specification	AIR-510F6	Info.	4 Nonrepro.	Recurring	Beginning 120 days after contract award and at 120-day intervals thereafter.
		LGR	Info.	5 "		
		NAVAIRSYSCOMREPCEN	Info.	3 "		
		NAVAIRSYSCOMREP PAC or LANT as applicable				
		NADC	Info.	3 "		
		NAEL SI	Info.	1 "		
		COMNAVAIRLANT	Info.	1 "		
		COMNAVAIRPAC	Info.	1 "		
		ASO	Info.	2 "		
3.4.44	Logbook	AIR-535	Accept.	1 Nonrepro.	Prelim.	At least 45 days prior to shipment of first target.
		NAVMISCEN	Info.	4 "		
		NADC	Info.	1 "		
		NAEL SI	Info.	1 "		
		with each target	Info.	1 "	Final	Ship with each target.
3.4.45	Demonstration Plan	AIR-535	Accept.	10 Nonrepro.	Nonrecurring	At least 90 days prior to first demonstration flight.
		NADC	Info.	3 "		
3.4.46	Demonstration Report	AIR-535	Accept.	6 Nonrepro.	Nonrecurring	Not later than 60 days after completion of tests and reduction of data.
		NADC	Info.	3 "		
3.4.47	Alteration Report Demonstration Targets	AIR-535	Accept.	3 Nonrepro.	Recurring	Not later than 10 days prior to target flight which incorporates the alteration.
		NAVMISCEN	Info.	2 "		
		NADC	Info.	1 "		
		NAEL SI	Info.	1 "		

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TABLE I (Cont)

Paragraph	Title	To Attn:	Action	Number and Kind of Copies	Type of Report	Submittal Date
3.4.48	Microfilm and Tabulating Cards.	NATSF	Accept.	1 Set Repro.	Recurring	As specified in the contract or purchase order.
3.4.49	Interchangeability Data	AIR-535 LGR AIR-4107	Accept. Info. Info.	3 Nonrepro. 1 " 1 "	Recurring	Not later than 30 days prior to system review. Revisions as program progresses.
3.4.50	Radar Reflectivity Report	AIR-535 NAVMISCEN NADC	Accept. Info. Info.	3 Nonrepro. 2 " 2 "	Nonrecurring	As specified by NAVAIR, but not later than 30 days prior to first flight.
3.4.51	Mission Planning and Operational Manual	AIR-535 NAVMISCEN	Accept. Info.	3 Nonrepro. 4 "	Recurring	
				Preliminary		90 days after award of contract.
		AIR-52043A NADC NAVMISCEN NATC	Accept. Info. Info. Info.	3 Nonrepro. 3 " 4 " 5 "	Final	Completion of contract.
3.4.53	Summary of Design Data	AIR-510F6 LGR NAVMISCEN NADC NAEL SI AIR-52043	Info. Accept. Info. Info. Info. Info.	4 Nonrepro. 1 " 1 " 3 " 1 " 1 "	Recurring	Every 90 days until completion of contract.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-D-8685B(AS) DATA AND TESTS, ENGINEERING: CONTRACT REQUIREMENTS FOR GUIDED MISSILE ~~TARGET SYSTEMS~~

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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